SOURCE-RESOLVED CHAOS EXPANSIONS FOR ITERATED BROWNIAN MOTION

RAMIRO FONTES

ABSTRACT. For the iterated Brownian motion $Z_t = X_{Y_t}$, with two-sided Brownian scenery X and independent Brownian clock Y, we study the occupation functional $F_T(\varphi) = \int_0^T \varphi(X_{Y_s})\,ds$, which by the occupation-time formula equals $\int_{\mathbb{R}} \varphi(X_x)\,L_T^x(Y)\,dx$, and expand it in the product Wiener chaos $\bigoplus_{m,n\geq 0} \mathcal{H}_m^X \otimes \mathcal{H}_n^Y$. We prove the source projection formula

$$\mathrm{Proj}_{m,n} F_T(\varphi) \;=\; \int_{\mathbb{R}} \mathrm{Proj}_m^X \varphi(X_x)\,\mathrm{Proj}_n^Y L_T^x(Y)\,dx, \qquad m,n\geq 0,$$

and obtain closed-form bidegree-$(1,1)$ and $(2,1)$ kernels for $F_T(\varphi)$, the variance identity $\mathrm{Var}(\mathrm{Proj}_{1,1}F_T(\varphi)) = \|K_T\|^2_{L^2(\mathbb{R}\times[0,T])}$ for an explicit K_T, and parity selection rules from the scenery Hermite components. On the joint isonormal we represent $\mathrm{Proj}_{1,1}F_T(\varphi) = I_2(\tilde{f}_T)$ and prove a non-asymptotic Nourdin–Peccati Stein bound for $\mathrm{Proj}_{1,1}F_T(\varphi)/\sigma_T$ against $N(0,1)$ via the 1-contraction $\tilde{f}_T \otimes_1 \tilde{f}_T$. As $T\to\infty$ with φ fixed and $\int_{\mathbb{R}} y\,\varphi(y)\,dy \neq 0$, this contraction and σ_T^2 scale at the same rate, obstructing a quantitative CLT through this route; convergence in the moment-zero subclass remains a conjecture. The product Wiener chaos and Malliavin machinery are standard; we use the occupation-time factorisation to obtain the source-resolved bidegree organisation, the closed-form low-bidegree kernels, the non-asymptotic Stein bound, and the $T\to\infty$ obstruction.

1. INTRODUCTION

Iterated Brownian motion (IBM) is the process

$$Z_t \;=\; X_{Y_t}, \qquad t\geq 0,$$

where $X = (X_x)_{x\in\mathbb{R}}$ is a two-sided standard Brownian motion (the *scenery*) and $Y = (Y_s)_{s\geq 0}$ is an independent standard Brownian motion (the *clock*). The process Z is non-Markovian, non-Gaussian, and has been studied as the canonical example of a *Brownian-time* process [6, 1]; it is associated with fourth-order parabolic and local-time PDE traces arising from $u(t,x) = \mathbb{E}[u_0(x+Z_t)]$ and related functionals; we return to this briefly in the Conclusion.

The functional $L^2(\Omega_X\times\Omega_Y)$ admits, by independence, the product Wiener chaos decomposition

$$L^2(\Omega_X\times\Omega_Y) \;=\; \bigoplus_{m,n\geq 0} \mathcal{H}_m^X \otimes \mathcal{H}_n^Y, \tag{1}$$

where $\mathcal{H}_m^X$ is the m-th Wiener chaos of the scenery and $\mathcal{H}_n^Y$ is the n-th Wiener chaos of the clock. Any square-integrable functional F of (X,Y) thus has a doubly-indexed expansion

$$F \;=\; \sum_{m,n\geq 0} F_{m,n}, \qquad F_{m,n}\in \mathcal{H}_m^X\otimes\mathcal{H}_n^Y.$$

The product chaos decomposition itself is well known; our point is that for IBM-type functionals the pair (m,n) carries information that the total-order chaos expansion (collapsing to $F = \sum_k \mathrm{Proj}_k F$ with $\mathrm{Proj}_k = \bigoplus_{m+n=k}\mathrm{Proj}_{m,n}$) conceals.

Date: June 8, 2026.

For IBM occupation functionals the source-factored representation makes the product-chaos bidegree computable. Composition entangles the two Gaussian spaces, but the entanglement has a clean integral representation: by the occupation-time formula, for $\varphi \in C_c^\infty(\mathbb{R})$,

$$F_T(\varphi) \;:=\; \int_0^T \varphi(X_{Y_s})\, ds \;=\; \int_{\mathbb{R}} \varphi(X_x)\, L_T^x(Y)\, dx, \tag{2}$$

where $L_T^x(Y)$ is the local time of the clock at level x up to time T. The right-hand side visibly factors the two sources: $x \mapsto \varphi(X_x)$ is a functional of the scenery alone, $x \mapsto L_T^x(Y)$ is a functional of the clock alone, and the spatial integral pairs them. The three resulting sectors, scenery innovations ($m \geq 1$, $n = 0$), clock or local-time innovations ($m = 0$, $n \geq 1$), and mixed scenery–clock interaction ($m \geq 1$, $n \geq 1$), are detected by the partial (source-specific) Malliavin derivatives D_X, D_Y and their adjoints δ_X, δ_Y on each Gaussian factor. These operators record sector detection, the sourcewise Stein/Meyer identity, and the double-counting correction for the mixed sector. The main result is the source projection formula

$$\mathrm{Proj}_{m,n} F_T(\varphi) \;=\; \int_{\mathbb{R}} \mathrm{Proj}_m^X \varphi(X_x)\, \mathrm{Proj}_n^Y L_T^x(Y)\, dx \qquad (m, n \geq 0),$$

proved in theorem 7.6.

Wiener chaos and Malliavin methods have already been applied to iterated Brownian motion. Khoshnevisan and Lewis [6] develop a stochastic calculus for Brownian motion on a Brownian fracture, of which IBM is a special case; Nourdin and Peccati [8] use chaos and Stein-type techniques to study weighted power variations of iterated Brownian motion. Chaos expansions of Brownian and fractional-Brownian local time go back at least to Nualart and Vives [11] and Hu and Øksendal [4]; chaos methods for related local-time functionals appear in Hu and Nualart's work on renormalised self-intersection local time for fractional Brownian motion [3]. Our functionals and emphasis are different: we study occupation functionals $F_T(\varphi) = \int_{\mathbb{R}} \varphi(X_x)\, L_T^x(Y)\, dx$ and use the spatial factorisation to organise the bidegree-(m, n) components of the product chaos. We obtain a source-resolved bidegree organisation of $F_T(\varphi)$ together with closed-form low-bidegree kernels and a sectorwise Stein/Meyer accounting.

The Malliavin calculus on the joint Gaussian space already expands F. The operators $D_X, D_Y, \delta_X, \delta_Y$ are the Malliavin derivative and Skorokhod integral applied to one Gaussian factor at a time. These are partial Malliavin derivatives in the natural sense. What the occupation-time factorisation buys us is the ability to compute the bidegree-(m, n) components of $F_T(\varphi)$, organise the centred functional into pure-scenery, pure-clock, and mixed sectors, and write down explicit kernels for the lowest mixed bidegrees. The mixed components $\mathcal{H}_m^X \otimes \mathcal{H}_n^Y$ with $m, n \geq 1$ carry the scenery–clock interaction, and the resulting picture supports a probabilistic interpretation of the fourth-order/local-time PDE traces of IBM.

1.1. **Main results.** We show:

(1) The general source projection formula

$$\mathrm{Proj}_{m,n} F_T(\varphi) \;=\; \int_{\mathbb{R}} \mathrm{Proj}_m^X \varphi(X_x)\, \mathrm{Proj}_n^Y L_T^x(Y)\, dx \qquad (m, n \geq 0),$$

which is the main result of the paper (theorem 7.6).

(2) Explicit closed-form bidegree-$(1, 1)$ mixed kernel for $F_T(\varphi)$ (theorem 7.3): $\mathrm{Proj}_{1,1} F_T(\varphi) = \delta_X \delta_Y K_T$ with $K_T \in L^2(\mathbb{R} \times [0, T])$ identified by direct product projection.

(3) Sector decomposition $F - \mathbb{E}[F] = F_X^{\mathrm{pure}} + F_Y^{\mathrm{pure}} + F_{\mathrm{mix}}$ with explicit double-counting correction for the naive sum of single-source Stein/Meyer identities (theorems 4.6 and 4.8).

(4) Variance identity at bidegree $(1,1)$ $\mathrm{Var}(\mathrm{Proj}_{1,1}F_T(\varphi)) = \|K_T\|^2_{L^2(\mathbb{R}\times[0,T])}$ (theorem 7.7); a direct corollary of the double Wiener–Itô isometry, with one possible numerical use as a source-resolved control variate left to future work.
(5) Parity selection rules: for φ even, $\mathrm{Proj}_{1,0}F_T = \mathrm{Proj}_{0,1}F_T = \mathrm{Proj}_{1,1}F_T = 0$; for φ odd, the entire pure-clock sector $\mathrm{Proj}_{0,n}F_T$, $n \geq 1$, vanishes (theorems 7.8 and 7.9).
(6) Explicit bidegree-$(2,1)$ mixed chaos (theorem 7.11), the first mixed bidegree containing the second scenery chaos.
(7) Non-asymptotic Nourdin–Peccati Stein bound for the normalised projection $G_T = \mathrm{Proj}_{1,1}F_T(\varphi)/\sigma_T$ against $N(0,1)$, via the 1-contraction of $\tilde{f}_T$ on the joint isonormal Gaussian space (theorem 8.3).
(8) Large-T same-rate growth: when $\varphi \in C_c^\infty(\mathbb{R})$ is fixed with $\int_{\mathbb{R}} y\,\varphi(y)\,dy \neq 0$, the variance σ_T^2 and the contraction norm $\|\tilde{f}_T \otimes_1 \tilde{f}_T\|$ both grow at rate T, so the Stein bound does not vanish as $T \to \infty$ (theorems 8.4 and 8.5); a quantitative CLT in the moment-zero subclass is conjectured (theorem 8.7).

Numerical verification of the source projection formula and the parity selection rules, together with a control-variate application of the interaction-energy identity, is left to future work.

2. Setup and notation

Let $(\Omega_X, \mathcal{F}^X, \mathbb{P}_X)$ carry a two-sided standard Brownian motion $X = (X_x)_{x\in\mathbb{R}}$ with $X_0 = 0$ and independent increments on $(-\infty, 0]$ and $[0,\infty)$, and let $(\Omega_Y, \mathcal{F}^Y, \mathbb{P}_Y)$ carry an independent standard Brownian motion $Y = (Y_s)_{s\geq 0}$. We work on the product space

$$(\Omega, \mathcal{F}, \mathbb{P}) \;=\; (\Omega_X \times \Omega_Y,\, \mathcal{F}^X \otimes \mathcal{F}^Y,\, \mathbb{P}_X \otimes \mathbb{P}_Y).$$

By $\mathbb{E}, \mathbb{E}_X, \mathbb{E}_Y$ we mean expectation with respect to $\mathbb{P}, \mathbb{P}_X, \mathbb{P}_Y$ respectively. Conditional expectations $\mathbb{E}_X[\cdot] \coloneqq \mathbb{E}[\cdot \mid \mathcal{F}^Y]$ and $\mathbb{E}_Y[\cdot] \coloneqq \mathbb{E}[\cdot \mid \mathcal{F}^X]$ will recur (note: $\mathbb{E}_X[\cdot]$ integrates out X, leaving a function of Y).

The associated isonormal Gaussian processes are denoted $W^X : \mathcal{H}_X \to L^2(\Omega_X)$ and $W^Y : \mathcal{H}_Y \to L^2(\Omega_Y)$, where $\mathcal{H}_X = L^2(\mathbb{R}, dx)$ and $\mathcal{H}_Y = L^2(\mathbb{R}_+, ds)$ in the natural identifications $X_x = W^X(\sigma_x)$ and $Y_s = W^Y(\mathbf{1}_{[0,s]})$, where the scenery Cameron–Martin kernel is

$$\sigma_x(\xi) \;\coloneqq\; \mathbf{1}_{(0,x)}(\xi) \,-\, \mathbf{1}_{(x,0)}(\xi), \qquad x, \xi \in \mathbb{R}, \tag{3}$$

writing $(0,x) = \emptyset$ when $x \leq 0$ and $(x,0) = \emptyset$ when $x \geq 0$, so that σ_x is supported between 0 and x with the appropriate two-sided sign. In particular $X_x = W^X(\sigma_x)$ is two-sided Brownian motion with $\langle \sigma_x, \sigma_{x'} \rangle_{\mathcal{H}_X} = \min(|x|, |x'|)$ when $xx' > 0$ and zero when $xx' < 0$.

Definition 2.1 (Iterated Brownian motion)**.** The iterated Brownian motion is the $\mathcal{F}$-measurable process

$$Z_t \;\coloneqq\; X_{Y_t}, \qquad t \geq 0.$$

For $T > 0$ and $x \in \mathbb{R}$, write $L_T^x(Y)$ for the (jointly continuous) local time of Y at level x up to time T, normalized so that the occupation-time formula holds in the form

$$\int_0^T g(Y_s)\,ds \;=\; \int_{\mathbb{R}} g(x)\, L_T^x(Y)\,dx \qquad \text{for nonnegative Borel } g. \tag{4}$$

We will repeatedly view the random field $x \mapsto L_T^x(Y)$ as a functional of Y alone, hence an element of $L^2(\Omega_Y; L^p(\mathbb{R}))$ for appropriate p; see [12, 2] for the local-time construction and Tanaka's formula.

Let $\varphi \in C_c^\infty(\mathbb{R})$. Define

$$F_T(\varphi) \;:=\; \int_0^T \varphi(Z_s)\,ds \;=\; \int_0^T \varphi(X_{Y_s})\,ds. \tag{5}$$

Applying (4) pointwise in ω_X, with $g(x) = \varphi(X_x(\omega_X))$, yields the central representation

$$F_T(\varphi) \;=\; \int_{\mathbb{R}} \varphi(X_x)\, L_T^x(Y)\,dx, \tag{6}$$

valid $\mathbb{P}$-a.s.; the joint measurability of $(\omega, x) \mapsto \varphi(X_x)\, L_T^x(Y)$ that turns this into an identity between genuine random variables, and that lets the right-hand side be read simultaneously as an ω-wise Lebesgue integral and as an $L^2(\Omega)$-valued Bochner integral, is established in theorem 5.2. Equation (6) is the central representation used throughout the paper: it factors the scenery randomness ($x \mapsto \varphi(X_x)$) from the clock randomness ($x \mapsto L_T^x(Y)$), pairing them through the spatial integral.

Throughout the paper we assume, unless stated otherwise:

(1) $\varphi \in C_c^\infty(\mathbb{R})$, so all integrals in (5)–(6) are absolutely convergent $\mathbb{P}$-a.s. and $F_T(\varphi) \in L^p(\Omega)$ for all $p \geq 1$ (using boundedness of φ and $\mathbb{E}[(L_T^x(Y))^p] < \infty$ uniformly on compact sets in x);
(2) $T > 0$ is fixed.

3. Product Wiener chaos

For each $m \geq 0$, let $\mathcal{H}_m^X \subset L^2(\Omega_X, \mathcal{F}^X, \mathbb{P}_X)$ be the m-th Wiener chaos of the scenery isonormal process W^X, and similarly $\mathcal{H}_n^Y \subset L^2(\Omega_Y, \mathcal{F}^Y, \mathbb{P}_Y)$ the n-th chaos of W^Y. Recall that

$$L^2(\Omega_X, \mathcal{F}^X, \mathbb{P}_X) \;=\; \bigoplus_{m\geq 0} \mathcal{H}_m^X, \qquad L^2(\Omega_Y, \mathcal{F}^Y, \mathbb{P}_Y) \;=\; \bigoplus_{n\geq 0} \mathcal{H}_n^Y,$$

and that every chaos space admits a unitary identification with the symmetric tensor power of the underlying Cameron–Martin space. By independence of X and Y, $L^2(\Omega, \mathcal{F}, \mathbb{P}) = L^2(\Omega_X, \mathbb{P}_X) \otimes L^2(\Omega_Y, \mathbb{P}_Y)$ (Hilbert tensor product). This factorization is compatible with the chaos decomposition on each factor, and yields:

Theorem 3.1 (Product Wiener chaos decomposition)**.** *There is an orthogonal decomposition*

$$L^2(\Omega, \mathcal{F}, \mathbb{P}) \;=\; \bigoplus_{m,n\geq 0} \mathcal{H}_m^X \otimes \mathcal{H}_n^Y. \tag{7}$$

For every $F \in L^2(\Omega)$ there is a unique sequence $(F_{m,n})_{m,n\geq 0}$ with $F_{m,n} \in \mathcal{H}_m^X \otimes \mathcal{H}_n^Y$ such that $F = \sum_{m,n\geq 0} F_{m,n}$ in L^2, and $\mathbb{E}[F^2] = \sum_{m,n\geq 0} \mathbb{E}[F_{m,n}^2]$.

Proof. We first make the tensor identification explicit, since this is the only point at which measurability enters. By construction $\mathcal{F} = \mathcal{F}^X \otimes \mathcal{F}^Y$ is exactly the product σ-algebra (no completion is taken) and $\mathbb{P} = \mathbb{P}_X \otimes \mathbb{P}_Y$. For $f \in L^2(\Omega_X)$ and $g \in L^2(\Omega_Y)$ the function

$$(f \otimes g)(\omega_X, \omega_Y) \;:=\; f(\omega_X)\, g(\omega_Y)$$

is $\mathcal{F}$-measurable, being the pointwise product of the $\mathcal{F}$-measurable maps $(\omega_X, \omega_Y) \mapsto f(\omega_X)$ and $(\omega_X, \omega_Y) \mapsto g(\omega_Y)$; and by Tonelli's theorem on the product measure,

$$\mathbb{E}[(f \otimes g)^2] \;=\; \int_{\Omega_X} f(\omega_X)^2 \Big(\int_{\Omega_Y} g(\omega_Y)^2\, \mathbb{P}_Y(d\omega_Y) \Big) \mathbb{P}_X(d\omega_X) \;=\; \|f\|_{L^2(\Omega_X)}^2\, \|g\|_{L^2(\Omega_Y)}^2. \tag{8}$$

Thus the bilinear map $(f, g) \mapsto f \otimes g$ induces a linear isometry $U\colon L^2(\Omega_X) \otimes L^2(\Omega_Y) \to L^2(\Omega)$ from the Hilbert tensor product into $L^2(\Omega)$. Its range is dense: for $A \in \mathcal{F}^X$ and $B \in \mathcal{F}^Y$ one has $U(\mathbf{1}_A \otimes \mathbf{1}_B) = \mathbf{1}_{A\times B}$, and the linear span of the rectangle indicators $\{\mathbf{1}_{A\times B} : A \in \mathcal{F}^X,\, B \in$

$\mathcal{F}^Y\}$ is dense in $L^2(\Omega,\mathcal{F},\mathbb{P})$. Indeed the measurable rectangles form a π-system generating $\mathcal{F}=\mathcal{F}^X\otimes\mathcal{F}^Y$, so by the monotone class theorem the bounded functions lying in the L^2-closure of the rectangle span—a vector space containing the rectangle indicators and stable under bounded monotone limits—exhaust the bounded $\mathcal{F}$-measurable functions, which are themselves L^2-dense. Hence U is onto and is the asserted unitary identification $L^2(\Omega)=L^2(\Omega_X)\otimes L^2(\Omega_Y)$ [5, §3.1]; it is precisely the independence of X and Y, i.e. $\mathbb{P}=\mathbb{P}_X\otimes\mathbb{P}_Y$, that underlies (8).

Each factor has its own Wiener chaos decomposition into closed subspaces, $L^2(\Omega_X)=\bigoplus_{m\geq 0}\mathcal{H}_m^X$ and $L^2(\Omega_Y)=\bigoplus_{n\geq 0}\mathcal{H}_n^Y$ [7, §1.1]. The Hilbert tensor product of two orthogonal direct sums of closed subspaces is the orthogonal direct sum of the pairwise (closed) tensor products, and U carries $\mathcal{H}_m^X\otimes\mathcal{H}_n^Y$ onto a closed subspace of $L^2(\Omega)$; this yields (7). The Parseval identity $\mathbb{E}[F^2]=\sum_{m,n}\mathbb{E}[F_{m,n}^2]$ and the uniqueness of the expansion follow from mutual orthogonality of the summands. □

Define the total-order projection

$$\mathrm{Proj}_k F \;\coloneqq\; \sum_{\substack{m,n\geq 0\\ m+n=k}} F_{m,n}, \qquad k\geq 0,$$

so that $F=\sum_k \mathrm{Proj}_k F$ is the chaos expansion that one would write if $L^2(\Omega)$ were treated as a single Gaussian space. Theorem 3.1 refines this:

$$\mathrm{Proj}_k F \;=\; \bigoplus_{m+n=k} \mathrm{Proj}_{m,n} F, \qquad \mathrm{Proj}_{m,n} F \coloneqq F_{m,n}.$$

The bidegree (m,n) records how many scenery innovations and how many clock innovations are present. Total chaos order $k=m+n$ collapses these. We also write

$$\mathrm{Proj}_X F \;\coloneqq\; \sum_{m\geq 0} F_{m,0} \;=\; \mathbb{E}_Y[F] \qquad \text{(scenery-only part)},$$

$$\mathrm{Proj}_Y F \;\coloneqq\; \sum_{n\geq 0} F_{0,n} \;=\; \mathbb{E}_X[F] \qquad \text{(clock-only part)},$$

where the conditional-expectation identifications follow from the fact that chaoses indexed by $(m,0)$ are $\mathcal{F}^X$-measurable and chaoses indexed by $(0,n)$ are $\mathcal{F}^Y$-measurable. The complementary part $F-\mathrm{Proj}_X F-\mathrm{Proj}_Y F+\mathbb{E}[F]$ collects the strictly mixed components $\bigoplus_{m,n\geq 1}\mathcal{H}_m^X\otimes\mathcal{H}_n^Y$.

4. Source-specific Malliavin derivatives and adjoints

The operators D_X, D_Y and their adjoints δ_X,δ_Y used in this section are the Malliavin derivative and Skorokhod integral on each Gaussian factor of $\Omega_X\times\Omega_Y$; see [7]. We use the term *source derivative* as shorthand for the Malliavin derivative restricted to one Gaussian factor; equivalent terminology is *partial Malliavin derivative*. The reason for naming them is bookkeeping: their bidegree action, the sourcewise Stein/Meyer identity, and the double-counting correction for the mixed sector are most easily stated in this notation. We define two partial Malliavin derivatives, one per source, by acting only on the corresponding factor of the product Gaussian space.

Definition 4.1 (Scenery derivative D_X)**.** Let $F\in L^2(\Omega)$ admit, $\mathbb{P}_Y$-a.s., a Malliavin-differentiable representative as a functional of X in the sense of the Malliavin calculus on $(\Omega_X,\mathcal{F}^X,\mathbb{P}_X)$. The *scenery derivative* is the $\mathcal{H}_X$-valued random field

$$(D_X F)(x) \;\coloneqq\; D_X^{(\mathrm{Mall.})} F(x), \qquad x\in\mathbb{R},$$

defined for $\mathbb{P}_Y$-a.e. ω_Y by applying the Malliavin derivative D^X on Ω_X to the section $\omega_X \mapsto F(\omega_X, \omega_Y)$. We denote its domain by $\mathbb{D}_X^{1,2} \subset L^2(\Omega)$.

Definition 4.2 (Clock derivative D_Y)**.** Symmetrically, for F in the appropriate domain $\mathbb{D}_Y^{1,2}$,

$$(D_Y F)(s) \;:=\; D_Y^{(\text{Mall.})} F(s), \qquad s \geq 0,$$

is the Malliavin derivative on Ω_Y applied to the section $\omega_Y \mapsto F(\omega_X, \omega_Y)$ for $\mathbb{P}_X$-a.e. ω_X.

Definition 4.3 (Adjoints δ_X, δ_Y)**.** Let δ_X (resp. δ_Y) be the Skorokhod integral on Ω_X (resp. Ω_Y), regarded as a partially-defined operator on $L^2(\Omega; \mathcal{H}_X)$ (resp. $L^2(\Omega; \mathcal{H}_Y)$) via $\langle \delta_X u, F\rangle_{L^2(\Omega)} = \mathbb{E}[\langle u, D_X F\rangle_{\mathcal{H}_X}]$ on the appropriate domains, and similarly for δ_Y.

The two derivatives act diagonally on the product chaos decomposition (7): D_X lowers the scenery index m by one and preserves n, while D_Y lowers n by one and preserves m.

Proposition 4.4 (Index lowering and raising)**.** *For $F_{m,n} \in \mathcal{H}_m^X \otimes \mathcal{H}_n^Y$ in the appropriate domains,*

$$D_X F_{m,n} \in \mathcal{H}_{m-1}^X \otimes \mathcal{H}_n^Y \otimes \mathcal{H}_X, \qquad D_Y F_{m,n} \in \mathcal{H}_m^X \otimes \mathcal{H}_{n-1}^Y \otimes \mathcal{H}_Y,$$

where $\mathcal{H}_{-1}^X = \mathcal{H}_{-1}^Y = \{0\}$ by definition. Conversely, δ_X raises m by one (preserving n) and δ_Y raises n by one (preserving m):

$$\delta_X :\; \mathcal{H}_m^X \otimes \mathcal{H}_n^Y \otimes \mathcal{H}_X \to \mathcal{H}_{m+1}^X \otimes \mathcal{H}_n^Y, \qquad \delta_Y :\; \mathcal{H}_m^X \otimes \mathcal{H}_n^Y \otimes \mathcal{H}_Y \to \mathcal{H}_m^X \otimes \mathcal{H}_{n+1}^Y.$$

The two pairs commute on smooth elements: $D_X D_Y = D_Y D_X$, and $\delta_X \delta_Y = \delta_Y \delta_X$ as densely defined operators (modulo domain conditions).

Proof. By bilinearity and density it suffices to verify the assertions on elementary tensors. Let $f_m \in \mathcal{H}_X^{\otimes m}$ and $g_n \in \mathcal{H}_Y^{\otimes n}$ be symmetric and consider $F_{m,n} = I_m^X(f_m)\, I_n^Y(g_n) \in \mathcal{H}_m^X \otimes \mathcal{H}_n^Y$, where I_m^X, I_n^Y are the multiple Wiener–Itô integrals on each factor. Conditional on the clock, $\omega_Y \mapsto I_n^Y(g_n)(\omega_Y)$ is a scalar; the single-source index-lowering identity for the Malliavin derivative on $(\Omega_X, \mathcal{F}^X, \mathbb{P}_X)$ [7, Prop. 1.2.1–1.2.2] gives, $\mathbb{P}_Y$-a.s.,

$$(D^X I_m^X(f_m))(\xi) \;=\; m\, I_{m-1}^X(f_m(\xi, \cdot)), \qquad \xi \in \mathbb{R}.$$

Hence

$$(D_X F_{m,n})(\xi) \;=\; m\, I_{m-1}^X(f_m(\xi,\cdot))\, I_n^Y(g_n) \;\in\; \mathcal{H}_{m-1}^X \otimes \mathcal{H}_n^Y \otimes \mathcal{H}_X,$$

where $\mathcal{H}_X = L^2(\mathbb{R}, dx)$ records the ξ slot. The clock case $D_Y F_{m,n} \in \mathcal{H}_m^X \otimes \mathcal{H}_{n-1}^Y \otimes \mathcal{H}_Y$ is symmetric. The operators extend from finite sums of such elementary tensors to the full domain $\mathbb{D}_X^{1,2}, \mathbb{D}_Y^{1,2}$ by closure, and the adjoint statements for δ_X, δ_Y follow from the duality relation in theorem 4.3 together with the corresponding single-source raising identity for the Skorokhod integral. For the commutation $D_X D_Y = D_Y D_X$ on elementary tensors:

$$D_X D_Y F_{m,n}(\xi, s) \;=\; D_X\big[\, n\, I_m^X(f_m)\, I_{n-1}^Y(g_n(s,\cdot))\,\big](\xi) \;=\; m\, n\, I_{m-1}^X(f_m(\xi,\cdot))\, I_{n-1}^Y(g_n(s,\cdot)),$$

and the same expression results from $D_Y D_X F_{m,n}$ by interchanging the two single-source chain rules, which act on the independent Gaussian factors and therefore commute. By density and closability, $D_X D_Y = D_Y D_X$ on smooth elements of the joint domain; $\delta_X \delta_Y = \delta_Y \delta_X$ follows by taking adjoints. □

Let $L_X = -\delta_X D_X$ and $L_Y = -\delta_Y D_Y$ be the Ornstein–Uhlenbeck generators on each factor. They act diagonally on the product chaos:

$$L_X\big|_{\mathcal{H}_m^X \otimes \mathcal{H}_n^Y} = -m \cdot \mathrm{Id}, \qquad L_Y\big|_{\mathcal{H}_m^X \otimes \mathcal{H}_n^Y} = -n \cdot \mathrm{Id}.$$

We define the source-specific pseudo-inverses by their explicit chaos action: for $F = \sum_{m,n\geq 0} F_{m,n} \in L^2(\Omega)$,

$$\Pi_X F \;\coloneqq\; \sum_{\substack{m\geq 1\\ n\geq 0}} \frac{1}{m}\, F_{m,n}, \qquad \Pi_Y F \;\coloneqq\; \sum_{\substack{m\geq 0\\ n\geq 1}} \frac{1}{n}\, F_{m,n}, \tag{9}$$

and zero on the kernel of the corresponding generator. Equivalently $\Pi_X = -L_X^{-1}$ on $(\ker L_X)^\perp$ and $\Pi_X|_{\ker L_X} = 0$, and analogously for Π_Y. With this normalization, the standard Stein/Meyer identity, applied source-by-source, reads

$$F - \mathbb{E}_X F \;=\; \delta_X(D_X\, \Pi_X F), \qquad F - \mathbb{E}_Y F \;=\; \delta_Y(D_Y\, \Pi_Y F), \tag{10}$$

where $\mathbb{E}_X[\cdot] = \mathbb{E}[\cdot \mid \mathcal{F}^Y]$ annihilates bidegrees with $m \geq 1$ and $\mathbb{E}_Y[\cdot] = \mathbb{E}[\cdot \mid \mathcal{F}^X]$ annihilates bidegrees with $n \geq 1$. The product chaos splits canonically into three orthogonal sectors.

Definition 4.5 (Source sectors)**.** For $F \in L^2(\Omega)$ with chaos expansion $F = \sum_{m,n\geq 0} F_{m,n}$, set

$$F_X^{\mathrm{pure}} \coloneqq \sum_{m\geq 1} F_{m,0}, \qquad F_Y^{\mathrm{pure}} \coloneqq \sum_{n\geq 1} F_{0,n}, \qquad F_{\mathrm{mix}} \coloneqq \sum_{m,n\geq 1} F_{m,n}.$$

Proposition 4.6 (Sector decomposition of the centred functional)**.** *For every $F \in L^2(\Omega)$,*

$$F - \mathbb{E}[F] \;=\; F_X^{\mathrm{pure}} \;+\; F_Y^{\mathrm{pure}} \;+\; F_{\mathrm{mix}}, \tag{11}$$

the three summands are mutually orthogonal in $L^2(\Omega)$, and

$$F_X^{\mathrm{pure}} = \mathbb{E}_Y F - \mathbb{E}[F], \quad F_Y^{\mathrm{pure}} = \mathbb{E}_X F - \mathbb{E}[F], \quad F_{\mathrm{mix}} = F - \mathbb{E}_X F - \mathbb{E}_Y F + \mathbb{E}[F]. \tag{12}$$

Proof. By independence, $\mathbb{E}_Y F = \sum_{m\geq 0} F_{m,0}$ and $\mathbb{E}_X F = \sum_{n\geq 0} F_{0,n}$, so the right-hand identities of (12) agree with theorem 4.5 after subtracting $\mathbb{E}[F] = F_{0,0}$. Equation (11) is the chaos-by-chaos identity $\sum_{(m,n)\neq(0,0)} F_{m,n}$, and orthogonality is inherited from theorem 3.1. □

The two single-source derivatives D_X, D_Y and the cross-derivative $D_X D_Y$ detect the three sectors of theorem 4.6 independently; they do not assemble into a single combined identity without an additional term accounting for the mixed sector, as theorem 4.8 below records.

Proposition 4.7 (Sector detection)**.** *For F in the appropriate domain:*

1. $F_X^{\mathrm{pure}} = \delta_X(D_X\, \Pi_X(\mathbb{E}_Y F))$*, equivalently the bidegree-$(\bullet, 0)$ component of $\delta_X(D_X \Pi_X F)$.*
2. $F_Y^{\mathrm{pure}} = \delta_Y(D_Y\, \Pi_Y(\mathbb{E}_X F))$*, equivalently the bidegree-$(0, \bullet)$ component of $\delta_Y(D_Y \Pi_Y F)$.*
3. $F_{\mathrm{mix}} = 0$ *if and only if all chaos kernels of $D_X D_Y F$ vanish; the bidegree $(1,1)$ part is*

$$F_{1,1} \;=\; \delta_X \delta_Y\, \mathbb{E}\big[D_X D_Y F\big],$$

and more generally each $F_{m,n}$ with $m, n \geq 1$ is recovered from the chaos kernel of $(D_X)^m (D_Y)^n F$.

Proof. For (1), set $G \coloneqq \mathbb{E}_Y F$. Then G is $\mathcal{F}^X$-measurable, hence $\mathbb{E}_X G = \mathbb{E}[G] = \mathbb{E}[F]$. Applying (10) on the scenery factor to G gives $G - \mathbb{E}_X G = \delta_X(D_X\, \Pi_X G)$, and the left-hand side is $G - \mathbb{E}[F] = \mathbb{E}_Y F - \mathbb{E}[F] = F_X^{\mathrm{pure}}$ by (12); the right-hand side has all bidegrees with $n = 0$, recovering the bidegree-$(\bullet, 0)$ component of $\delta_X(D_X\, \Pi_X F)$ since $\mathbb{E}_Y$ commutes with the scenery operators D_X, δ_X, Π_X. Item (2) is symmetric in X, Y. For (3), $D_X D_Y$ lowers both bidegree indices by one and preserves all others; thus $\mathbb{E}[D_X D_Y F] \in \mathcal{H}_X \otimes \mathcal{H}_Y$ is exactly the chaos kernel of $F_{1,1} \in \mathcal{H}_1^X \otimes \mathcal{H}_1^Y$, and applying the joint adjoint $\delta_X \delta_Y$ inverts the index lowering, giving $F_{1,1} = \delta_X \delta_Y\, \mathbb{E}[D_X D_Y F]$. Iterating $D_X^m D_Y^n$ recovers each $F_{m,n}$ with $m, n \geq 1$ from the same chaos kernel. □

Remark 4.8. Identity (10) gives, on bidegree (m,n), $\delta_X(D_X\Pi_X F) = F_{m,n}$ for $m \geq 1$ and zero for $m = 0$, so $\delta_X(D_X\Pi_X F) = F_X^{\text{pure}} + F_{\text{mix}}$. Symmetrically $\delta_Y(D_Y\Pi_Y F) = F_Y^{\text{pure}} + F_{\text{mix}}$, and adding gives

$$\delta_X(D_X\Pi_X F) + \delta_Y(D_Y\Pi_Y F) \;=\; F_X^{\text{pure}} + F_Y^{\text{pure}} + 2\,F_{\text{mix}} \;=\; (F - \mathbb{E}[F]) + F_{\text{mix}}.$$

The mixed sector is counted twice, which is why we use the sectorwise identities of theorem 4.7 rather than a single combined formula. Equivalently,

$$F - \mathbb{E}[F] \;=\; \delta_X(D_X\Pi_X F) + \delta_Y(D_Y\Pi_Y F) - F_{\text{mix}},$$

with F_{mix} as in (12).

5. The occupation functional

We collect the source-specific derivative formulas for

$$F_T(\varphi) \;=\; \int_0^T \varphi(X_{Y_s})\,ds \;=\; \int_{\mathbb{R}} \varphi(X_x)\,L_T^x(Y)\,dx, \qquad \varphi \in C_c^\infty(\mathbb{R}),$$

that we use later: the scenery derivative $D_X F_T$ is ordinary (theorem 5.3); the unmollified clock derivative $D_Y F_T$ is singular and is handled through a mollified scenery (theorem 5.4); and the finite-ε mixed derivative $D_X D_Y F_T^\varepsilon$ has a closed form (theorem 5.5) whose bidegree-$(1,1)$ kernel is reconciled with the direct-projection result of theorem 7.3 in theorem 7.5. The bidegree-(m,n) projections of $F_T(\varphi)$ are obtained in section 7 by direct product projection, not via the partial Malliavin derivatives; the present section provides operator-level context.

Lemma 5.1 (Moments of $F_T(\varphi)$)**.** *For every $p \geq 1$, $F_T(\varphi) \in L^p(\Omega)$, with $\mathbb{E}[|F_T(\varphi)|^p] \leq T^p\|\varphi\|_\infty^p$.*

Proof. Bound $|F_T(\varphi)| \leq T\|\varphi\|_\infty$ pointwise. □

Lemma 5.2 (Spatial Bochner integrability of the product)**.** *For $\varphi \in C_c^\infty(\mathbb{R})$ and $T > 0$, the map $x \mapsto \varphi(X_x)\,L_T^x(Y)$ from $\mathbb{R}$ to $L^2(\Omega)$ is strongly measurable and Bochner integrable, with*

$$\big\|\varphi(X_x)\,L_T^x(Y)\big\|_{L^2(\Omega)}^2 \;=\; \mathbb{E}[\varphi(X_x)^2]\cdot\mathbb{E}[L_T^x(Y)^2], \tag{13}$$

and there is a constant $C_{\varphi,T} > 0$, finite on bounded T-intervals, such that

$$\big\|\varphi(X_x)\,L_T^x(Y)\big\|_{L^2(\Omega)} \;\leq\; C_{\varphi,T}\,(1+|x|)^{-1/4}\,e^{-x^2/(4T)}, \qquad x \in \mathbb{R}. \tag{14}$$

In particular $\int_{\mathbb{R}} \varphi(X_x)\,L_T^x(Y)\,dx \in L^2(\Omega)$ and equals $F_T(\varphi)$ $\mathbb{P}$-a.s. by Fubini. Moreover, for every bounded linear operator S on $L^2(\Omega)$ the map $x \mapsto S\big(\varphi(X_x)\,L_T^x(Y)\big)$ is strongly measurable and Bochner integrable, and

$$S\int_{\mathbb{R}} \varphi(X_x)\,L_T^x(Y)\,dx \;=\; \int_{\mathbb{R}} S\big(\varphi(X_x)\,L_T^x(Y)\big)\,dx. \tag{15}$$

Proof. Write $\Psi(x) \coloneqq \varphi(X_x)\,L_T^x(Y) \in L^2(\Omega)$. Bochner integrability is the conjunction of strong measurability and integrability of the norm [5, §3.1]; we verify the two in turn and then deduce the operator statement.

Joint measurability. Both coordinate fields admit jointly measurable versions. The scenery X has a version with continuous sample paths $x \mapsto X_x$, so $(\omega_X, x) \mapsto X_x(\omega_X)$ is $\mathcal{F}^X \otimes \mathcal{B}(\mathbb{R})$-measurable; and the local time $(\omega_Y, x) \mapsto L_T^x(Y)(\omega_Y)$ has a version jointly continuous in x, hence $\mathcal{F}^Y \otimes \mathcal{B}(\mathbb{R})$-measurable [12, Ch. VI]. As φ is continuous, the map $(\omega, x) \mapsto \Psi(x)(\omega) = \varphi(X_x(\omega_X))\,L_T^x(Y)(\omega_Y)$ is $\mathcal{F} \otimes \mathcal{B}(\mathbb{R})$-measurable.

Strong measurability. We show that $x \mapsto \Psi(x)$ is continuous from $\mathbb{R}$ into $L^2(\Omega)$; since $L^2(\Omega)$ is separable, continuity implies strong (Bochner) measurability. For $x, x' \in \mathbb{R}$ decompose

$$\Psi(x) - \Psi(x') \;=\; (\varphi(X_x) - \varphi(X_{x'}))\, L_T^x(Y) \;+\; \varphi(X_{x'})\, (L_T^x(Y) - L_T^{x'}(Y)).$$

By independence of X and Y the $L^2(\Omega)$-norm of each summand factors, so by the triangle inequality

$$\begin{aligned}\|\Psi(x) - \Psi(x')\|_{L^2(\Omega)} \;\le\; &\big\|\varphi(X_x) - \varphi(X_{x'})\big\|_{L^2(\Omega_X)}\, \big\|L_T^x(Y)\big\|_{L^2(\Omega_Y)} \\ &+\; \|\varphi\|_\infty\, \big\|L_T^x(Y) - L_T^{x'}(Y)\big\|_{L^2(\Omega_Y)}.\end{aligned}$$

As $x' \to x$ the first Ω_X-factor tends to 0, because φ is Lipschitz and bounded and $\mathbb{E}[(X_x - X_{x'})^2] = |x - x'| \to 0$ (dominated convergence on $(\varphi(X_x) - \varphi(X_{x'}))^2 \le \min\{4\|\varphi\|_\infty^2, \|\varphi'\|_\infty^2 (X_x - X_{x'})^2\}$); the local-time factor $\|L_T^x(Y)\|_{L^2(\Omega_Y)}$ is bounded uniformly in x by the moment estimate below; and the last Ω_Y-factor tends to 0 by the L^2-continuity in the space variable of Brownian local time, which follows from the joint continuity together with the uniform moment bound $\sup_x \mathbb{E}[L_T^x(Y)^4] < \infty$ on bounded T-intervals [12, Ch. VI]. Hence Ψ is $L^2(\Omega)$-continuous and strongly measurable.

Norm integrability. Equation (13) is independence of X and Y. For (14), take $A > 0$ with $\operatorname{supp}\varphi \subseteq [-A, A]$. Then $\varphi(X_x)^2 \le \|\varphi\|_\infty^2$ and $\mathbb{E}[\varphi(X_x)^2] \le \|\varphi\|_\infty^2\, \mathbb{P}(|X_x| \le A) = \|\varphi\|_\infty^2\, [2\Phi(A/\sqrt{|x|}) - 1]$, with $X_x \sim N(0, |x|)$. As $|x| \to \infty$, $2\Phi(A/\sqrt{|x|}) - 1 \sim 2A\,(2\pi|x|)^{-1/2}$, so $\mathbb{E}[\varphi(X_x)^2] = O(|x|^{-1/2})$. For the local-time moment, the standard Gaussian transition density bound gives, e.g. [12, Ch. VI], $\mathbb{E}[L_T^x(Y)^2] \le c\, e^{-x^2/(2T)}$ for a constant depending on T on bounded intervals (in particular $\sup_x \mathbb{E}[L_T^x(Y)^2] \le c < \infty$, the bound used above). Combining, $\big\|\varphi(X_x) L_T^x(Y)\big\|_2^2 = O(|x|^{-1/2}\, e^{-x^2/(2T)})$, which absorbs into (14). The right-hand side is integrable on $\mathbb{R}$, so $\int_{\mathbb{R}} \|\Psi(x)\|_{L^2(\Omega)}\, dx < \infty$.

Conclusion. Strong measurability and norm integrability give, by Bochner's theorem, that Ψ is Bochner integrable. The joint measurability of $(\omega, x) \mapsto \Psi(x)(\omega)$ together with $\int_{\mathbb{R}} \|\Psi(x)\|_2\, dx < \infty$ licenses Fubini's theorem, so the $L^2(\Omega)$-valued Bochner integral $\int_{\mathbb{R}} \Psi(x)\, dx$ coincides $\mathbb{P}$-a.s. with the ω-wise scalar integral $\int_{\mathbb{R}} \varphi(X_x)\, L_T^x(Y)\, dx$, which by the occupation-time formula (4) applied to the bounded Borel function $g(\cdot) = \varphi(X_\cdot(\omega_X))$ along the path $Y(\omega_Y)$ equals $F_T(\varphi)$; this is (6). Finally, if S is a bounded linear operator on $L^2(\Omega)$, then $x \mapsto S\Psi(x)$ is strongly measurable (the composition of a strongly measurable map with a bounded operator) and $\|S\Psi(x)\|_2 \le \|S\|\, \|\Psi(x)\|_2$ is integrable, so $S\Psi$ is Bochner integrable; the commutation (15) is then the defining property of the Bochner integral, that bounded operators pass through it [5, §3.1]. □

For the scenery derivative, the clock plays the role of a *random measure* $\mu_T(dx) \coloneqq L_T^x(Y)\, dx$ on $\mathbb{R}$, independent of X, giving a direct computation through the source-factored representation.

Proposition 5.3 (Scenery derivative)**.** *$F_T(\varphi) \in \mathbb{D}_X^{1,2}$, and*

$$(D_X F_T(\varphi))(\xi) \;=\; \int_{\mathbb{R}} \varphi'(X_x)\, \sigma_x(\xi)\, L_T^x(Y)\, dx, \qquad \xi \in \mathbb{R}, \tag{16}$$

with σ_x as in (3)*; the two-sided sign is already built into σ_x.*

Proof. Conditional on Y, the random measure $\mu_T(dx) = L_T^x(Y)\, dx$ is deterministic. The map $\omega_X \mapsto \int_{\mathbb{R}} \varphi(X_x(\omega_X))\, \mu_T(dx)$ is then a smooth functional of the scenery, and the Malliavin derivative on Ω_X commutes with the spatial integral by dominated convergence (using compact support of φ and integrability of $L_T^x(Y)$). The formula (16) then follows from $D^X(X_x)(\xi) = \sigma_x(\xi)$

(the two-sided sign is already built into σ_x from (3)); see [7] for the scenery-Malliavin chain rule. □

The clock derivative is more delicate than the scenery derivative because of the irregularity of the scenery in the spatial variable. A naive chain rule on the clock factor produces an ill-defined object; we must approach $D_Y F_T$ through a smoothing of X and analyze the limit.

The obstruction is the spatial roughness of the scenery. For a smooth function $g \in C_b^1(\mathbb{R})$, the Malliavin chain rule on Ω_Y gives

$$D_s^Y \int_0^T g(Y_u)\, du \;=\; \int_s^T g'(Y_u)\, du, \qquad s \in [0,T]. \tag{17}$$

For $F_T(\varphi) = \int_0^T \varphi(X_{Y_u})\, du$, one would like to apply (17) with $g(x) = \varphi(X_x)$. But for $\mathbb{P}_X$-a.e. ω_X the random function $x \mapsto \varphi(X_x)$ is only $1/2 - \eta$ Hölder continuous and not differentiable in x, so (17) cannot be applied directly: the right-hand side is not even defined.

Spatial mollification repairs the chain rule. Let $\rho \in C_c^\infty(\mathbb{R})$ be a non-negative kernel with $\int \rho = 1$, and set $\rho_\varepsilon(x) = \varepsilon^{-1}\rho(x/\varepsilon)$. Define the mollified scenery

$$X_x^\varepsilon \;\coloneqq\; (X * \rho_\varepsilon)(x) \;=\; \int_{\mathbb{R}} X_y\, \rho_\varepsilon(x-y)\, dy,$$

a C^∞ field in x for $\mathbb{P}_X$-a.e. ω_X, and the mollified occupation functional

$$F_T^\varepsilon(\varphi) \;\coloneqq\; \int_0^T \varphi(X_{Y_u}^\varepsilon)\, du.$$

Proposition 5.4 (Clock derivative of the mollified functional)**.** *For each $\varepsilon > 0$, $F_T^\varepsilon(\varphi) \in \mathbb{D}_Y^{1,2}$, and*

$$\big(D_Y F_T^\varepsilon(\varphi)\big)(s) \;=\; \int_s^T \varphi'(X_{Y_u}^\varepsilon)\, \partial_x X_{Y_u}^\varepsilon\, du, \qquad s \in [0,T]. \tag{18}$$

Proof. Conditional on X, the field $g^\varepsilon(x) = \varphi(X_x^\varepsilon)$ is C^1 in x with bounded derivatives on compact sets. Apply (17) with $g = g^\varepsilon$ and use $(g^\varepsilon)'(y) = \varphi'(X_y^\varepsilon)\cdot \partial_x X_y^\varepsilon$. Take expectation over X; integrability follows from boundedness of φ' and the deterministic kernel ρ_ε. □

The $\varepsilon \to 0$ limit must be read distributionally. Equation (18) contains the factor $\partial_x X^\varepsilon = X * \rho_\varepsilon'$, whose pointwise variance is of order ε^{-1}, so the unmollified $D_Y F_T(\varphi)$ is not pointwise defined on Ω_Y due to the spatial roughness of the scenery; we do not need it, since theorem 7.3 obtains the bidegree-$(1,1)$ projection by direct product projection in Section 7. The chaos projections $\mathrm{Proj}_{m,n} D_Y F_T(\varphi)$ are nonetheless well-defined.

Combining the two derivatives gives the operator that detects scenery–clock interaction. We compute it on the mollified functional, where every step is justified at finite ε, and identify the part that is stable under $\varepsilon \to 0$.

We first record the Cameron–Martin kernel of the mollified scenery. For two-sided Brownian scenery X, the Malliavin derivative on Ω_X acts as

$$D_X X_z(\xi) \;=\; \mathbf{1}_{(0,z)}(\xi) \;-\; \mathbf{1}_{(z,0)}(\xi), \qquad z, \xi \in \mathbb{R},$$

where $(0,z) = \emptyset$ for $z \le 0$ and analogously for $(z,0)$. Define the *mollified Cameron–Martin kernel* by

$$H_\varepsilon(y,\xi) \;\coloneqq\; D_X X_y^\varepsilon(\xi) \;=\; \int_{\mathbb{R}} \big[\, \mathbf{1}_{(0,z)}(\xi) - \mathbf{1}_{(z,0)}(\xi)\,\big]\, \rho_\varepsilon(y-z)\, dz, \qquad y, \xi \in \mathbb{R}. \tag{19}$$

For each $\varepsilon > 0$, H_ε is jointly smooth, bounded (by 1), and supported, in the ξ variable, on a neighborhood of the interval between 0 and y of width $O(\varepsilon)$. The chain rule for D_X together with the commutation of ∂_y with D_X give

$$D_X X_y^\varepsilon(\xi) \;=\; H_\varepsilon(y,\xi), \qquad D_X(\partial_y X_y^\varepsilon)(\xi) \;=\; \partial_y H_\varepsilon(y,\xi). \tag{20}$$

Proposition 5.5 (Mixed derivative of the mollified functional)**.** *For each $\varepsilon > 0$, $F_T^\varepsilon(\varphi) \in \mathbb{D}_{X,Y}^{2,2}$, and*

$$(D_X D_Y F_T^\varepsilon(\varphi))(\xi,s) \;=\; \int_s^T \Big[\, \varphi''(X_{Y_u}^\varepsilon)\, \partial_y X_{Y_u}^\varepsilon\, H_\varepsilon(Y_u,\xi) + \;\varphi'(X_{Y_u}^\varepsilon)\, \partial_y H_\varepsilon(Y_u,\xi)\Big]\, du, \qquad \xi \in \mathbb{R},\; s \in [0,T]. \tag{21}$$

The bidegree-$(1,1)$ kernel $K_T^\varepsilon := \mathbb{E}[D_X D_Y F_T^\varepsilon(\varphi)]$ extracted from (21) lies in $L^2(\mathbb{R}\times[0,T])$ for each $\varepsilon > 0$; its $\varepsilon \to 0$ reconciliation with the kernel of theorem 7.3 is established in theorem 7.5.

Proof. At fixed $\varepsilon > 0$, the integrand of (18) is $\varphi'(X_{Y_u}^\varepsilon)\, \partial_y X_{Y_u}^\varepsilon$, a product of two scenery-differentiable factors. Apply D_X via the product rule on $(\Omega_X, \mathcal{F}^X, \mathbb{P}_X)$, conditional on Y:

$$D_X\big(\varphi'(X_{Y_u}^\varepsilon)\big)(\xi) \;=\; \varphi''(X_{Y_u}^\varepsilon)\, D_X X_{Y_u}^\varepsilon(\xi) \;=\; \varphi''(X_{Y_u}^\varepsilon)\, H_\varepsilon(Y_u,\xi)$$

by the standard scenery chain rule and (20), and

$$D_X\big(\partial_y X_{Y_u}^\varepsilon\big)(\xi) \;=\; \partial_y\, D_X X_{Y_u}^\varepsilon(\xi) \;=\; \partial_y H_\varepsilon(Y_u,\xi)$$

by commutation of ∂_y with D_X and (20). Interchange of D_X with the u-integral in (18) is justified by boundedness of the scenery derivatives at finite ε and dominated convergence. Combining the two pieces under the u-integral yields (21). The L^2-membership of $K_T^\varepsilon = \mathbb{E}[D_X D_Y F_T^\varepsilon]$ at finite ε follows from boundedness of $H_\varepsilon, \partial_y H_\varepsilon, \varphi', \varphi''$ on the support of φ together with the local-time moment estimate of theorem 5.2. The $\varepsilon \to 0$ reconciliation is established separately in theorem 7.5. □

In summary, theorem 5.3 holds without mollification, while theorems 5.4 and 5.5 hold at finite ε; the unmollified $D_Y F_T$ is a generalised object; and the $\varepsilon \to 0$ reconciliation at the bidegree-$(1,1)$ kernel is theorem 7.5.

6. First-order source split

We now describe the source-resolved first-order chaos decomposition of $F_T(\varphi)$ and determine what is gained over the total-first-chaos projection.

Recall $\mathrm{Proj}_{m,n}$ from Section 3. The source-resolved first-order content of $F_T(\varphi)$ is

$$\mathrm{Proj}_{1,0} F_T(\varphi) \;\in\; \mathcal{H}_1^X \otimes \mathcal{H}_0^Y \quad \text{(scenery innovation)},$$

$$\mathrm{Proj}_{0,1} F_T(\varphi) \;\in\; \mathcal{H}_0^X \otimes \mathcal{H}_1^Y \quad \text{(clock innovation)}.$$

The total first-chaos projection, $\mathrm{Proj}_1 F_T = \mathrm{Proj}_{1,0} F_T + \mathrm{Proj}_{0,1} F_T$, sums these.

Proposition 6.1 (Scenery first-order projection)**.** *With $F_T = F_T(\varphi)$, $\mathrm{Proj}_{1,0} F_T$ is the first scenery chaos of*

$$\mathrm{Proj}_X F_T \;=\; \mathbb{E}_Y[F_T] \;=\; \int_{\mathbb{R}} \mathbb{E}_Y[L_T^x(Y)]\, \varphi(X_x)\, dx,$$

and equals

$$\mathrm{Proj}_{1,0} F_T \;=\; \int_{\mathbb{R}} p_T(x)\, \mathbb{E}[\varphi'(X_x)]\, X_x\, dx \;=\; \int_{\mathbb{R}} p_T(x)\, \mathbb{E}[\varphi'(X_x)]\, W^X(\sigma_x)\, dx, \tag{22}$$

where $p_T(x) \coloneqq \mathbb{E}[L_T^x(Y)]$ is the deterministic mean local time density (Gaussian transition kernel integrated over $[0,T]$).

Proof. By Fubini and integrability of $L_T^x(Y)$, $\mathrm{Proj}_X F_T = \mathbb{E}_Y F_T = \int_{\mathbb{R}} p_T(x)\,\varphi(X_x)\,dx$, with $p_T(x) = \mathbb{E}[L_T^x(Y)]$. Project onto $\mathcal{H}_1^X$: for each fixed x, the random variable $\varphi(X_x) \in L^2(\sigma(X_x))$ has Hermite expansion

$$\varphi(X_x) \;=\; \mathbb{E}[\varphi(X_x)] + \mathbb{E}[\varphi'(X_x)]\,X_x + R, \qquad R \in \bigoplus_{m\geq 2} \mathcal{H}_m^X,$$

by Stein's identity $\mathbb{E}[X_x\,\varphi(X_x)] = |x|\,\mathbb{E}[\varphi'(X_x)]$ together with $\|X_x\|^2 = |x|$. Hence $\mathrm{Proj}_1^X \varphi(X_x) = \mathbb{E}[\varphi'(X_x)]\,X_x = \mathbb{E}[\varphi'(X_x)]\,W^X(\sigma_x)$ using $X_x = W^X(\sigma_x)$ from (3). The deterministically weighted field $x \mapsto p_T(x)\,\varphi(X_x)$ is strongly measurable and Bochner integrable from $\mathbb{R}$ to $L^2(\Omega_X)$ by the continuity-and-decay argument of theorem 5.2: it is $L^2(\Omega_X)$-continuous (since $x \mapsto \varphi(X_x)$ is, by the Lipschitz bound used there, and p_T is continuous) and norm integrable (since p_T is bounded with Gaussian decay $p_T(x) = O(e^{-x^2/(2T)})$ and $\|\varphi(X_x)\|_2 \leq \|\varphi\|_\infty$). The orthogonal projection $\mathrm{Proj}_{1,0}$ on $L^2(\Omega_X)$ is a bounded operator (operator norm 1), so it commutes with the spatial Bochner integral $\int_{\mathbb{R}} p_T(x)\,\varphi(X_x)\,dx$, and taking the projection inside yields (22). □

Proposition 6.2 (Clock first-order projection)**.**

$$\mathrm{Proj}_{0,1} F_T \;=\; \int_{\mathbb{R}} \mathbb{E}_X[\varphi(X_x)] \cdot \mathrm{Proj}_{0,1}\big(L_T^x(Y)\big)\,dx, \tag{23}$$

and the first clock chaos of the local time at level x is the Wiener integral

$$\mathrm{Proj}_{0,1}\big(L_T^x(Y)\big) \;=\; \int_0^T k_T(x,s)\,dY_s, \qquad k_T(x,s) \;\coloneqq\; \mathrm{erf}\Big(\tfrac{x}{\sqrt{2s}}\Big) - \mathrm{erf}\Big(\tfrac{x}{\sqrt{2T}}\Big), \tag{24}$$

for $s \in (0,T)$. Equivalently, $\mathrm{Proj}_{0,1}\big(L_T^x(Y)\big) = -\mathrm{erf}\big(\tfrac{x}{\sqrt{2T}}\big)\,Y_T + \int_0^T \mathrm{erf}\big(\tfrac{x}{\sqrt{2s}}\big)\,dY_s$.

Proof. By independence, $\mathrm{Proj}_Y F_T = \mathbb{E}_X F_T = \int_{\mathbb{R}} \mathbb{E}_X[\varphi(X_x)]\,L_T^x(Y)\,dx$, so (23) reduces to projecting $L_T^x(Y)$ onto $\mathcal{H}_1^Y$. Apply Tanaka's formula $L_T^x(Y) = |Y_T - x| - |x| - \int_0^T \mathrm{sgn}(Y_s - x)\,dY_s$ and project each term, using $\mathrm{Proj}_{\mathcal{H}_1^Y} G = W^Y(\mathbb{E}[D^Y G])$ and $\mathbb{E}[\mathrm{sgn}(Y_s - x)] = -\mathrm{erf}(x/\sqrt{2s})$:

$$\begin{aligned}
\mathrm{Proj}_{0,1}|Y_T - x| \;&=\; W^Y\big(\mathbb{E}[\mathrm{sgn}(Y_T - x)]\,\mathbf{1}_{[0,T]}\big) \;=\; -\mathrm{erf}\Big(\tfrac{x}{\sqrt{2T}}\Big)\,Y_T,\\
\mathrm{Proj}_{0,1}|x| \;&=\; 0,\\
\mathrm{Proj}_{0,1}\int_0^T \mathrm{sgn}(Y_s - x)\,dY_s \;&=\; \int_0^T \mathbb{E}[\mathrm{sgn}(Y_s - x)]\,dY_s \;=\; -\int_0^T \mathrm{erf}\Big(\tfrac{x}{\sqrt{2s}}\Big)\,dY_s,
\end{aligned}$$

where the last identity uses that the first chaos of an Itô integral $\int u_s\,dY_s$ with adapted u_s is $\int \mathbb{E}[u_s]\,dY_s$. Combining the three pieces with the signs of Tanaka's formula yields (24). The interchange of $\mathrm{Proj}_{0,1}$ with the spatial integral is again theorem 5.2: the field $x \mapsto \mathbb{E}_X[\varphi(X_x)]\,L_T^x(Y) = \mathbb{E}[\varphi(X_x)]\,L_T^x(Y)$ is strongly measurable and Bochner integrable into $L^2(\Omega_Y)$ (a deterministic multiple of $L_T^x(Y)$, to which the continuity-and-decay argument of theorem 5.2 applies), and $\mathrm{Proj}_{0,1}$ is bounded, so it passes inside the integral. The projected field $x \mapsto \mathbb{E}[\varphi(X_x)]\,\mathrm{Proj}_{0,1} L_T^x(Y) = \mathbb{E}[\varphi(X_x)]\,W^Y(k_T(x,\cdot)\mathbf{1}_{[0,T]})$ is $L^2(\Omega_Y)$-continuous in x (the isometry W^Y composed with the $L^2([0,T])$-continuous $x \mapsto k_T(x,\cdot)$, which is bounded by 2 and continuous on $\mathbb{R} \times (0,T]$) and absolutely integrable in x, using the decay $\|k_T(x,\cdot)\|_{L^2([0,T])} \leq C\sqrt{T}\,e^{-x^2/(2T)}$ for $|x| \geq \sqrt{T}$ from theorem 7.4; the chaos order is preserved by orthogonality. □

Remark 6.3 (Properties of k_T). $k_T(-x,s) = -k_T(x,s)$, so the kernel is odd in x as expected from the symmetry of Y around 0; $k_T(x,T) = 0$ (the integrand vanishes at the upper endpoint, reflecting the cancellation between the $|Y_T - x|$ endpoint contribution and the running Tanaka integral); and $k_T(0,s) = 0$, consistent with $L_T^0(Y)$ being an even functional of Y and hence orthogonal to $\mathcal{H}_1^Y$.

By orthogonality of $\mathcal{H}_1^X \otimes \mathcal{H}_0^Y$ and $\mathcal{H}_0^X \otimes \mathcal{H}_1^Y$ in the product chaos, the first-order projection $\mathrm{Proj}_1 F_T = \mathrm{Proj}_{1,0} F_T + \mathrm{Proj}_{0,1} F_T$ admits the variance decomposition $\mathrm{Var}(\mathrm{Proj}_1 F_T) = \mathrm{Var}(\mathrm{Proj}_{1,0} F_T) + \mathrm{Var}(\mathrm{Proj}_{0,1} F_T)$, which the total first-chaos projection alone does not record. The remainder $F_T - \mathbb{E}[F_T] - \mathrm{Proj}_{1,0} F_T - \mathrm{Proj}_{0,1} F_T$ lies in $\bigoplus_{m,n\geq 1} \mathcal{H}_m^X \otimes \mathcal{H}_n^Y$ and carries the scenery–clock interaction analysed in section 7.

7. Mixed chaos and scenery–clock interaction

Define the strictly mixed subspace

$$\mathcal{H}_{\mathrm{mix}} \;:=\; \bigoplus_{m\geq 1,\, n\geq 1} \mathcal{H}_m^X \otimes \mathcal{H}_n^Y \;\subset\; L^2(\Omega).$$

A functional $F \in L^2(\Omega)$ has zero projection onto $\mathcal{H}_{\mathrm{mix}}$ if and only if it can be decomposed (in L^2) as a sum of an $\mathcal{F}^X$-measurable and an $\mathcal{F}^Y$-measurable functional. In other words, $\mathcal{H}_{\mathrm{mix}}$ is the space of genuine scenery–clock interaction.

Proposition 7.1 (Mixed sector vs. tensor decomposition)**.** *For $F \in L^2(\Omega)$, the following are equivalent:*

1. $\mathrm{Proj}_{m,n} F = 0$ *for all $m, n \geq 1$;*
2. *there exist $G \in L^2(\Omega_X)$ and $H \in L^2(\Omega_Y)$ such that $F = G + H$ a.s. (mod constants).*

Proof. The implication (2)⇒(1) is immediate from $G \in \bigoplus_m \mathcal{H}_m^X \otimes \mathcal{H}_0^Y$ and $H \in \bigoplus_n \mathcal{H}_0^X \otimes \mathcal{H}_n^Y$. For (1)⇒(2), set $G = \mathbb{E}_Y F = \mathrm{Proj}_X F$ and $H = F - G$, and verify that all chaoses of H have $m = 0$ using (1) and the L^2-decomposition. □

The mixed second derivative $D_X D_Y$ is the operator that lowers both indices by one. Its smallest non-trivial output is the bidegree $(1,1)$ component, the first scenery–clock interaction chaos.

Proposition 7.2 ($D_X D_Y$ detects $(1,1)$)**.** *For $F \in L^2(\Omega)$ in the appropriate domain,*

$$\mathbb{E}\big[D_X D_Y F\big] \;\in\; \mathcal{H}_X \otimes \mathcal{H}_Y$$

is the integral kernel of the bilinear part of F in the sense that

$$\mathrm{Proj}_{1,1} F \;=\; \delta_X \delta_Y\Big(\mathbb{E}\big[D_X D_Y F\big]\Big),$$

and $\mathrm{Proj}_{1,1} F = 0$ if and only if $\mathbb{E}[D_X D_Y F] = 0$ in $\mathcal{H}_X \otimes \mathcal{H}_Y$.

Proof. Expand $F = \sum_{m,n\geq 0} F_{m,n}$ in the product chaos (7), with $F_{m,n} = I_m^X \otimes I_n^Y(\tilde{f}_{m,n})$ for symmetric kernels $\tilde{f}_{m,n} \in \mathcal{H}_X^{\otimes m} \otimes \mathcal{H}_Y^{\otimes n}$. By theorem 4.4, applied first on the scenery factor and then on the clock factor,

$$D_X D_Y F_{m,n}(\xi, s) \;=\; mn\, I_{m-1}^X \otimes I_{n-1}^Y(\tilde{f}_{m,n}(\xi, \cdot, s, \cdot)), \quad \xi \in \mathcal{H}_X,\; s \in \mathcal{H}_Y,$$

with the right-hand side zero whenever $m = 0$ or $n = 0$. Taking expectation $\mathbb{E}[\cdot]$ annihilates every chaos term except the order-zero one $(m-1, n-1) = (0,0)$, so

$$\mathbb{E}\big[D_X D_Y F\big](\xi, s) \;=\; \tilde{f}_{1,1}(\xi, s),$$

which lies in $\mathcal{H}_X \otimes \mathcal{H}_Y$. Applying $\delta_X \delta_Y$ returns $I_1^X \otimes I_1^Y(\tilde{f}_{1,1}) = F_{1,1} = \mathrm{Proj}_{1,1} F$. The "if and only if" statement follows from the double Wiener–Itô isometry $\|\delta_X \delta_Y g\|_{L^2(\Omega)} = \|g\|_{\mathcal{H}_X \otimes \mathcal{H}_Y}$, which makes $\delta_X \delta_Y$ injective on $\mathcal{H}_X \otimes \mathcal{H}_Y$. □

For $F_T = F_T(\varphi) = \int_{\mathbb{R}} \varphi(X_x) L_T^x(Y)\, dx$, the mixed sector is non-trivial. The $(1,1)$ component admits a closed-form description via direct product-chaos projection, bypassing the singular unmollified $D_Y F_T$.

Theorem 7.3 (Bidegree $(1,1)$ of the occupation functional)**.** *Let $\varphi \in C_c^\infty(\mathbb{R})$ and $T > 0$. Then $F_T(\varphi) \in L^2(\Omega)$ has bidegree-$(1,1)$ projection*

$$\mathrm{Proj}_{1,1} F_T(\varphi) \;=\; \int_{\mathbb{R}} \mathbb{E}[\varphi'(X_x)]\, X_x \int_0^T k_T(x,s)\, dY_s\, dx, \tag{25}$$

in $\mathcal{H}_1^X \otimes \mathcal{H}_1^Y$, where $k_T(x,s) = \mathrm{erf}(\frac{x}{\sqrt{2s}}) - \mathrm{erf}(\frac{x}{\sqrt{2T}})$ is the kernel of (24). Equivalently, with $\sigma_x(\xi) \coloneqq \mathbf{1}_{(0,x)}(\xi) - \mathbf{1}_{(x,0)}(\xi)$ so that $X_x = W^X(\sigma_x)$,

$$\mathrm{Proj}_{1,1} F_T(\varphi) \;=\; \delta_X \delta_Y\, K_T, \qquad K_T(\xi,s) \;\coloneqq\; \int_{\mathbb{R}} \mathbb{E}[\varphi'(X_x)]\, \sigma_x(\xi)\, k_T(x,s)\, dx, \tag{26}$$

with $K_T \in L^2(\mathbb{R} \times [0,T])$ and $\delta_X \delta_Y K_T$ the double Wiener–Itô integral.

Proof. For $G \in L^2(\Omega_X)$ with chaos expansion $G = \sum_{m \geq 0} G_m$ and $H \in L^2(\Omega_Y)$ with $H = \sum_{n \geq 0} H_n$, independence of X and Y gives, in $L^2(\Omega) = L^2(\Omega_X) \otimes L^2(\Omega_Y)$, $GH = \sum_{m,n \geq 0} G_m H_n$ with $G_m H_n \in \mathcal{H}_m^X \otimes \mathcal{H}_n^Y$. Hence

$$\mathrm{Proj}_{1,1}(GH) \;=\; \mathrm{Proj}_1^X G \cdot \mathrm{Proj}_1^Y H. \tag{27}$$

We proceed to the per-x projections. Fix $x \in \mathbb{R} \setminus \{0\}$ (the formulas below depend on $|x| > 0$ through the Hermite normalization; the singleton $\{0\}$ is Lebesgue-null in the spatial Bochner integral appearing below and we extend the integrand arbitrarily there). Apply (27) with $G = \varphi(X_x)$, $H = L_T^x(Y)$. We claim that

$$\mathrm{Proj}_1^X \varphi(X_x) \;\in\; \mathrm{span}(X_x). \tag{28}$$

The random variable $\varphi(X_x)$ is a Borel function of the single Gaussian $X_x \sim N(0, |x|)$ (with $X_x = W^X(\sigma_x)$), so $\varphi(X_x) \in L^2(\sigma(X_x))$. The chaos decomposition of $L^2(\sigma(X_x))$ inside $L^2(\Omega_X)$ is the Hermite expansion in $X_x/\sqrt{|x|}$: every $f(X_x) \in L^2(\sigma(X_x))$ has the form $f(X_x) = \sum_{m \geq 0} c_m(x)\, h_m(X_x/\sqrt{|x|})$, with $h_m(X_x/\sqrt{|x|}) \in \mathcal{H}_m^X$. The first-chaos summand $c_1(x)\, X_x/\sqrt{|x|}$ lies in $\mathrm{span}(X_x) \subset \mathcal{H}_1^X$, while the remaining $m \neq 1$ summands lie in chaoses orthogonal to $\mathcal{H}_1^X$; hence (28) holds. The projection onto a one-dimensional subspace is given by the standard formula. By Stein's lemma for $X_x \sim N(0,|x|)$, $\mathbb{E}[X_x\, f(X_x)] = |x|\, \mathbb{E}[f'(X_x)]$, applied with $f = \varphi$, the inner product is $\mathbb{E}[\varphi(X_x)\, X_x] = |x|\, \mathbb{E}[\varphi'(X_x)]$, hence

$$\mathrm{Proj}_1^X \varphi(X_x) \;=\; \frac{\mathbb{E}[\varphi(X_x) X_x]}{\mathbb{E}[X_x^2]}\, X_x \;=\; \mathbb{E}[\varphi'(X_x)]\, X_x.$$

By theorem 6.2, $\mathrm{Proj}_1^Y L_T^x(Y) = \int_0^T k_T(x,s)\, dY_s$. Combining,

$$\mathrm{Proj}_{1,1}\big(\varphi(X_x)\, L_T^x(Y)\big) \;=\; \mathbb{E}[\varphi'(X_x)]\, X_x \int_0^T k_T(x,s)\, dY_s. \tag{29}$$

To pass from this per-x identity to one for $F_T(\varphi)$, apply theorem 5.2: the map $x \mapsto \varphi(X_x)\, L_T^x(Y)$ is strongly measurable and Bochner integrable from $\mathbb{R}$ to $L^2(\Omega)$, with Gaussian decay in $|x|$, and the spatial integral equals $F_T(\varphi)$ almost surely. Since $\mathrm{Proj}_{1,1}$ is bounded on $L^2(\Omega)$, the operator statement of theorem 5.2 with $S = \mathrm{Proj}_{1,1}$ guarantees that the projected

field is itself strongly measurable and Bochner integrable and, by (15), that $\mathrm{Proj}_{1,1}$ commutes with the spatial integral:

$$\mathrm{Proj}_{1,1}F_T(\varphi) \;=\; \int_{\mathbb{R}} \mathrm{Proj}_{1,1}\big(\varphi(X_x)\, L_T^x(Y)\big)\, dx \tag{30}$$

in $L^2(\Omega)$.

Substituting (29) into (30) yields (25). Using $X_x = W^X(\sigma_x)$ and $\int_0^T k_T(x,s)\, dY_s = W^Y(k_T(x,\cdot)\,\mathbf{1}_{[0,T]})$, the integral $\int_{\mathbb{R}} \mathbb{E}[\varphi'(X_x)]\, W^X(\sigma_x)\, W^Y(k_T(x,\cdot))\, dx$ is the double Wiener–Itô integral of the kernel K_T in (26), which lies in $L^2(\mathbb{R}\times[0,T])$ by theorem 7.4 below. □

Lemma 7.4 (Stein-bound kernel is L^2)**.** *For $\varphi \in C_c^\infty(\mathbb{R})$ and $T > 0$, the kernel K_T defined by* (26) *satisfies $K_T \in L^2(\mathbb{R}\times[0,T])$. Moreover there is a constant $C(\varphi)$ depending only on φ such that*

$$\|K_T\|^2_{L^2(\mathbb{R}\times[0,T])} \;\le\; C(\varphi)\, T\,(1+\log T)^2 \qquad (T \ge 1). \tag{31}$$

Proof. We first establish the pointwise bound

$$|a(x)| \;\le\; C(\varphi)\,(1+|x|)^{-3/2}, \qquad x \in \mathbb{R}, \tag{32}$$

for some constant $C(\varphi)$. On $|x| \le 1$, $|a(x)| \le \|\varphi'\|_\infty$ gives the bound trivially. For $|x| \ge 1$, expand the $N(0,|x|)$ density in inverse powers of $|x|$ as in the derivation of (53): the leading term vanishes by $\int_{\mathbb{R}} \varphi'(y)\, dy = 0$ (compact support of φ), and the $|x|^{-3/2}$ coefficient is c_φ, with remainder $O(|x|^{-5/2})$. Hence $|a(x)| \le C(\varphi)\,|x|^{-3/2}$ for $|x| \ge 1$; combining both regimes gives (32).

Apply Minkowski's integral inequality to the spatial integral defining K_T:

$$\|K_T\|_{L^2(\mathbb{R}\times[0,T])} \;\le\; \int_{\mathbb{R}} |a(x)|\, \big\|\sigma_x(\cdot)\, k_T(x,\cdot)\big\|_{L^2(\mathbb{R}\times[0,T])}\, dx.$$

Since σ_x depends only on ξ and $k_T(x,\cdot)$ only on s, the inner norm factors:

$$\big\|\sigma_x(\cdot)\, k_T(x,\cdot)\big\|_{L^2(\mathbb{R}\times[0,T])} \;=\; \|\sigma_x\|_{L^2(\mathbb{R})}\, \|k_T(x,\cdot)\|_{L^2([0,T])} \;=\; \sqrt{|x|}\, \|k_T(x,\cdot)\|_{L^2([0,T])},$$

using $\|\sigma_x\|^2_{L^2(\mathbb{R})} = |x|$.

For the temporal norm: the uniform bound $|k_T(x,s)| \le 2$ gives $\|k_T(x,\cdot)\|_{L^2([0,T])} \le 2\sqrt{T}$ for all x. For $|x| \ge \sqrt{T}$, the tail estimate $1 - \mathrm{erf}(z) \le e^{-z^2}/(z\sqrt{\pi})$ yields the sharper bound $|k_T(x,s)| \le C\, e^{-x^2/(2T)}$ uniformly in $s \in [0,T]$, so $\|k_T(x,\cdot)\|_{L^2([0,T])} \le C\sqrt{T}\, e^{-x^2/(2T)}$ in this regime.

Combining with (32),

$$\begin{aligned}
\|K_T\|_{L^2(\mathbb{R}\times[0,T])} \;&\le\; C(\varphi) \int_{\mathbb{R}} (1+|x|)^{-3/2} \sqrt{|x|}\, \|k_T(x,\cdot)\|_{L^2([0,T])}\, dx \\
&\le\; 2C(\varphi)\sqrt{T} \int_{|x|\le\sqrt{T}} (1+|x|)^{-3/2}\sqrt{|x|}\, dx \;+\; C'\sqrt{T} \int_{|x|>\sqrt{T}} |x|^{-1}\, e^{-x^2/(2T)}\, dx.
\end{aligned}$$

The integrand of the first integral is $O(|x|^{-1})$ for large $|x|$ within the cutoff, giving $O(\log T)$ as $T \to \infty$. The second integral is bounded uniformly in $T \ge 1$ by a constant. Hence

$$\|K_T\|^2_{L^2(\mathbb{R}\times[0,T])} \;\le\; C(\varphi)^2\, T\,(1+\log T)^2$$

for some constant $C(\varphi)$ and all $T \ge 1$. In particular $K_T \in L^2(\mathbb{R}\times[0,T])$. □

Proposition 7.5 (Reconciliation of the mollified $D_X D_Y$ kernel)**.** *For each $\varepsilon > 0$ let $F_T^\varepsilon(\varphi) = \int_0^T \varphi(X^\varepsilon_{Y_u})\,du$ as in theorem 5.4, and define the deterministic bidegree-$(1,1)$ kernel*

$$K_T^\varepsilon \;:=\; \mathbb{E}\big[D_X D_Y F_T^\varepsilon(\varphi)\big] \;\in\; L^2(\mathbb{R}\times[0,T]),$$

with $D_X D_Y F_T^\varepsilon$ given by theorem 5.5. Let K_T be the kernel of theorem 7.3. Then

$$K_T^\varepsilon \;\xrightarrow[\varepsilon\to 0]{}\; K_T \qquad \text{in } L^2(\mathbb{R}\times[0,T]). \tag{33}$$

Proof. We work at the level of the chaos projections, without computing K_T^ε pointwise. For each $\varepsilon > 0$, $F_T^\varepsilon(\varphi) \in \mathbb{D}^{2,2}_{X,Y}$ by theorem 5.5, and theorem 7.2 applied to $G = F_T^\varepsilon(\varphi)$ gives

$$\mathrm{Proj}_{1,1} F_T^\varepsilon(\varphi) \;=\; \delta_X\delta_Y\,\mathbb{E}\big[D_X D_Y F_T^\varepsilon(\varphi)\big] \;=\; \delta_X\delta_Y K_T^\varepsilon, \tag{34}$$

while theorem 7.3 gives $\mathrm{Proj}_{1,1} F_T(\varphi) = \delta_X\delta_Y K_T$.

Next we establish $L^2(\Omega)$ convergence of the functionals themselves. For a non-negative kernel $\rho_\varepsilon \in C_c^\infty(\mathbb{R})$ with $\int \rho_\varepsilon = 1$ and supported in $[-\varepsilon,\varepsilon]$, write $X_x^\varepsilon - X_x = \int_{\mathbb{R}} (X_y - X_x)\,\rho_\varepsilon(x-y)\,dy$ and apply Cauchy–Schwarz:

$$\mathbb{E}\big[(X_x^\varepsilon - X_x)^2\big] \;\le\; \int_{\mathbb{R}} \mathbb{E}\big[(X_y - X_x)^2\big]\,\rho_\varepsilon(x-y)\,dy \;=\; \int_{\mathbb{R}} |y-x|\,\rho_\varepsilon(x-y)\,dy \;\le\; \varepsilon,$$

where we used $\mathbb{E}[(X_y - X_x)^2] = |y-x|$ for the two-sided scenery and $\mathrm{supp}(\rho_\varepsilon) \subseteq [-\varepsilon,\varepsilon]$. The bound is uniform in $x \in \mathbb{R}$. Since $\varphi \in C_c^\infty(\mathbb{R})$ is $\|\varphi'\|_\infty$-Lipschitz, Cauchy–Schwarz on du gives

$$\mathbb{E}\big[|F_T^\varepsilon(\varphi) - F_T(\varphi)|^2\big] \;\le\; T\,\|\varphi'\|_\infty^2 \int_0^T \mathbb{E}\big[(X^\varepsilon_{Y_u} - X_{Y_u})^2\big]\,du \;\le\; T^2\,\|\varphi'\|_\infty^2\,\varepsilon \;\to\; 0,$$

where independence of X and Y allows conditioning on Y_u to inherit the uniform bound.

Boundedness of $\mathrm{Proj}_{1,1}$ on $L^2(\Omega)$ (operator norm at most 1) lifts this convergence to the projection:

$$\big\|\mathrm{Proj}_{1,1} F_T^\varepsilon(\varphi) - \mathrm{Proj}_{1,1} F_T(\varphi)\big\|_{L^2(\Omega)} \;\le\; \big\|F_T^\varepsilon(\varphi) - F_T(\varphi)\big\|_{L^2(\Omega)} \;\to\; 0. \tag{35}$$

The double Wiener–Itô integral $\delta_X\delta_Y \colon L^2(\mathbb{R}\times[0,T]) \to \mathcal{H}_1^X \otimes \mathcal{H}_1^Y$ is an isometry: by independence of X and Y and the single-source Itô isometries, $\mathbb{E}\big[(W^X(f)W^Y(g))^2\big] = \|f\|^2_{\mathcal{H}_X}\|g\|^2_{\mathcal{H}_Y}$, which extends by bilinearity and density to $\|\delta_X\delta_Y K\|_{L^2(\Omega)} = \|K\|_{L^2(\mathbb{R}\times[0,T])}$ for every $K \in L^2(\mathbb{R}\times[0,T])$. Combining the chaos identity (34) with the projection convergence (35),

$$\|K_T^\varepsilon - K_T\|_{L^2(\mathbb{R}\times[0,T])} \;=\; \|\delta_X\delta_Y(K_T^\varepsilon - K_T)\|_{L^2(\Omega)} \;=\; \big\|\mathrm{Proj}_{1,1}F_T^\varepsilon(\varphi) - \mathrm{Proj}_{1,1}F_T(\varphi)\big\|_{L^2(\Omega)} \;\to\; 0,$$

which is (33). □

7.1. **Source projection at general bidegree.** The Bochner-integrability and projection-interchange argument of theorem 7.3 extends without modification to general bidegree (m,n). We record its three direct consequences.

Theorem 7.6 (Source projection formula)**.** *Let $\varphi \in C_c^\infty(\mathbb{R})$ and $T > 0$. For every $m, n \ge 0$,*

$$\mathrm{Proj}_{m,n} F_T(\varphi) \;=\; \int_{\mathbb{R}} \mathrm{Proj}_m^X \varphi(X_x)\;\cdot\;\mathrm{Proj}_n^Y L_T^x(Y)\,dx, \tag{36}$$

in $L^2(\Omega)$, the integral converging as a Bochner integral with values in $\mathcal{H}_m^X \otimes \mathcal{H}_n^Y$.

Proof. The argument follows the same lines as the proof of theorem 7.3, with $(1,1)$ replaced by (m,n).

By independence of X and Y and the tensor identification $L^2(\Omega) = L^2(\Omega_X) \otimes L^2(\Omega_Y)$, every product GH with $G \in L^2(\Omega_X)$ and $H \in L^2(\Omega_Y)$ admits the expansion $GH = \sum_{m',n' \geq 0} G_{m'} H_{n'}$ with $G_{m'} H_{n'} \in \mathcal{H}^X_{m'} \otimes \mathcal{H}^Y_{n'}$, hence

$$\mathrm{Proj}_{m,n}(GH) \;=\; \mathrm{Proj}^X_m G \cdot \mathrm{Proj}^Y_n H. \tag{37}$$

Applying (37) with $G = \varphi(X_x)$, $H = L^x_T(Y)$ gives the per-x identity

$$\mathrm{Proj}_{m,n}\big(\varphi(X_x)\, L^x_T(Y)\big) \;=\; \mathrm{Proj}^X_m \varphi(X_x) \;\cdot\; \mathrm{Proj}^Y_n L^x_T(Y). \tag{38}$$

To promote this to a global identity for $F_T(\varphi)$ we invoke the measurability furnished by theorem 5.2. The map $\mathrm{Proj}_{m,n} \colon L^2(\Omega) \to \mathcal{H}^X_m \otimes \mathcal{H}^Y_n$ is a bounded linear operator (orthogonal projection, operator norm at most 1), so the operator statement of theorem 5.2, with $S = \mathrm{Proj}_{m,n}$, applies: the projected field $x \mapsto \mathrm{Proj}_{m,n}\big(\varphi(X_x)\, L^x_T(Y)\big)$ is strongly measurable and Bochner integrable, and by (15)

$$\mathrm{Proj}_{m,n} F_T(\varphi) \;=\; \mathrm{Proj}_{m,n} \int_{\mathbb{R}} \varphi(X_x)\, L^x_T(Y)\, dx \;=\; \int_{\mathbb{R}} \mathrm{Proj}_{m,n}\big(\varphi(X_x)\, L^x_T(Y)\big)\, dx.$$

(The norm domination $\big\|\mathrm{Proj}_{m,n}\big(\varphi(X_x) L^x_T(Y)\big)\big\|_{L^2(\Omega)} \;=\; \big\|\mathrm{Proj}^X_m \varphi(X_x)\, \mathrm{Proj}^Y_n L^x_T(Y)\big\|_{L^2(\Omega)} \leq \big\|\varphi(X_x)\, L^x_T(Y)\big\|_{L^2(\Omega)}$, by contractivity of $\mathrm{Proj}^X_m, \mathrm{Proj}^Y_n$, recovers the integrability of the norm directly, but it is the strong measurability of the projected field, not merely this bound, that is supplied by theorem 5.2.) Combining with the per-x identity (38) yields (36). The output integral takes values in $\mathcal{H}^X_m \otimes \mathcal{H}^Y_n$ since each integrand value does and $\mathcal{H}^X_m \otimes \mathcal{H}^Y_n$ is closed in $L^2(\Omega)$. □

Corollary 7.7 (Interaction energy at bidegree $(1,1)$)**.** *With K_T the kernel of* (26)*, which lies in $L^2(\mathbb{R} \times [0,T])$ by theorem 7.4,*

$$\mathrm{Var}\big(\mathrm{Proj}_{1,1} F_T(\varphi)\big) \;=\; \|K_T\|^2_{L^2(\mathbb{R}\times[0,T])}. \tag{39}$$

Proof. Every element of $\mathcal{H}^X_m \otimes \mathcal{H}^Y_n$ with $(m,n) \neq (0,0)$ has zero mean (by orthogonality with constants), so $\mathrm{Var}(\mathrm{Proj}_{1,1} F_T) = \|\mathrm{Proj}_{1,1} F_T\|_2^2$. The double Wiener–Itô integral $\delta_X \delta_Y \colon L^2(\mathbb{R} \times [0,T]) \to \mathcal{H}^X_1 \otimes \mathcal{H}^Y_1$ is an isometry: by independence and the single-source isometries,

$$\mathbb{E}\big[\big(W^X(f)\, W^Y(g)\big)^2\big] \;=\; \mathbb{E}[W^X(f)^2] \cdot \mathbb{E}[W^Y(g)^2] \;=\; \|f\|^2_{\mathcal{H}_X}\, \|g\|^2_{\mathcal{H}_Y},$$

which extends by bilinearity and density to $\|\delta_X \delta_Y K\|_2^2 = \|K\|^2_{L^2(\mathbb{R}\times[0,T])}$ for every $K \in L^2(\mathbb{R} \times [0,T])$. Apply with $K = K_T$. □

The two-sided scenery satisfies $X_x \overset{d}{=} -X_x$ for every $x \in \mathbb{R}$, so expectations of odd functions of X_x vanish. This triggers two parity selection rules.

Proposition 7.8 (φ even)**.** *If $\varphi \colon \mathbb{R} \to \mathbb{R}$ is even, then*

$$\mathrm{Proj}_{1,0} F_T(\varphi) \;=\; 0, \qquad \mathrm{Proj}_{0,1} F_T(\varphi) \;=\; 0, \qquad \mathrm{Proj}_{1,1} F_T(\varphi) \;=\; 0.$$

Proof. Since φ is even, φ' is odd, and $X_x \sim N(0, |x|)$ is symmetric, so $\mathbb{E}[\varphi'(X_x)] = 0$ for every $x \in \mathbb{R}$. The integrand of $\mathrm{Proj}_{1,0} F_T$ in (22) contains $\mathbb{E}[\varphi'(X_x)]$ as a factor, so $\mathrm{Proj}_{1,0} F_T = 0$; the integrand of $\mathrm{Proj}_{1,1} F_T$ in (25) contains the same factor, so $\mathrm{Proj}_{1,1} F_T = 0$.

For $\mathrm{Proj}_{0,1} F_T$, apply theorem 7.6 with $(m,n) = (0,1)$:

$$\mathrm{Proj}_{0,1} F_T(\varphi) \;=\; \int_{\mathbb{R}} \mathbb{E}[\varphi(X_x)] \Big(\int_0^T k_T(x,s)\, dY_s \Big)\, dx.$$

With φ even and X_x symmetric, $\mathbb{E}[\varphi(X_x)]$ depends only on $|x|$, hence is even in x. The kernel $k_T(x,s) = \operatorname{erf}(x/\sqrt{2s}) - \operatorname{erf}(x/\sqrt{2T})$ is odd in x, since erf is odd. The spatial integrand is therefore odd in x as a random variable indexed by x, so the Bochner integral over $\mathbb{R}$ vanishes. □

Proposition 7.9 (φ odd: pure-clock sector vanishes)**.** *If $\varphi\colon \mathbb{R} \to \mathbb{R}$ is odd, then $\mathbb{E}[\varphi(X_x)] = 0$ for every $x \in \mathbb{R}$, and*

$$\operatorname{Proj}_{0,n} F_T(\varphi) \;=\; 0 \qquad \text{for every } n \geq 1.$$

Proof. For φ odd,

$$\mathbb{E}[\varphi(X_x)] \;=\; \int_{\mathbb{R}} \varphi(y)\,(2\pi|x|)^{-1/2}\, e^{-y^2/(2|x|)}\, dy \;=\; 0,$$

since the integrand is the product of an odd function with the even Gaussian density. By theorem 7.6 with $m = 0$,

$$\operatorname{Proj}_{0,n} F_T(\varphi) \;=\; \int_{\mathbb{R}} \mathbb{E}[\varphi(X_x)] \cdot \operatorname{Proj}_n^Y L_T^x(Y)\, dx \;=\; 0$$

for every $n \geq 1$. □

Corollary 7.10 (Generalized parity rule)**.** *For φ even, $\operatorname{Proj}_{m,n} F_T(\varphi) = 0$ for every odd $m \geq 1$ and every $n \geq 0$. For φ odd, $\operatorname{Proj}_{m,n} F_T(\varphi) = 0$ for every even $m \geq 0$ and every $n \geq 0$.*

Proof. By the Hermite parity of h_m: for φ even, $\operatorname{Proj}_m^X \varphi(X_x) = 0$ for every odd m, and dually for φ odd. Apply theorem 7.6. □

Iterating, $\operatorname{Proj}_{m,n} F_T(\varphi)$ for general $m, n \geq 1$ corresponds to the m-fold scenery iterated integral of the n-fold clock iterated integral of an explicit kernel built from the joint statistics of X and $L_T^\bullet(Y)$. The point is that the bidegree (m,n) counts the order of scenery–clock interaction:

- $(m,0)$: pure scenery (clock contributes only through the mean local time density);
- $(0,n)$: pure clock (scenery contributes only through expected values $\mathbb{E}[\varphi(X_x)]$);
- (m,n), $m,n \geq 1$: m-th order scenery fluctuation interacting with n-th order clock fluctuation.

This is the source-resolved chaos content of F_T. The total-order expansion sums these.

To illustrate concretely, we compute the bidegree $(2,1)$ projection, the first mixed bidegree containing a non-trivial second scenery chaos.

Theorem 7.11 (Bidegree $(2,1)$ of the occupation functional)**.** *Let $\varphi \in C_c^\infty(\mathbb{R})$ and $T > 0$. Then*

$$\operatorname{Proj}_{2,1} F_T(\varphi) \;=\; \int_{\mathbb{R}} \tfrac{1}{2}\, \mathbb{E}[\varphi''(X_x)]\, (X_x^2 - |x|) \int_0^T k_T(x,s)\, dY_s\, dx, \tag{40}$$

in $\mathcal{H}_2^X \otimes \mathcal{H}_1^Y$, the integral converging as a Bochner integral.

Proof. Apply theorem 7.6 with $(m,n) = (2,1)$. The first clock chaos is $\operatorname{Proj}_1^Y L_T^x(Y) = \int_0^T k_T(x,s)\, dY_s$ by theorem 6.2. For the second scenery chaos, the random variable $\varphi(X_x) \in L^2(\sigma(X_x))$ admits the Hermite expansion

$$\varphi(X_x) \;=\; \sum_{m \geq 0} \frac{\mathbb{E}[\varphi^{(m)}(X_x)]}{m!}\, h_m(X_x; |x|),$$

where $h_m(z;\sigma^2)$ is the Hermite polynomial of degree m orthogonal with respect to $N(0,\sigma^2)$, in the unnormalised form $h_2(z;\sigma^2) = z^2 - \sigma^2$ with $\|h_m(\cdot;\sigma^2)\|_2^2 = m!\,\sigma^{2m}$ (see [5, 7] for the

corresponding chaos normalisation), and the coefficients are fixed by the iterated Stein identity $\mathbb{E}[\varphi(X_x)\, h_m(X_x;|x|)] = |x|^m\, \mathbb{E}[\varphi^{(m)}(X_x)]$. Since $h_m(W^X(\sigma_x);|x|) \in \mathcal{H}_m^X$, the m-th summand equals $\mathrm{Proj}_m^X \varphi(X_x)$. For $m = 2$ this gives

$$\mathrm{Proj}_2^X \varphi(X_x) \;=\; \tfrac{1}{2}\, \mathbb{E}[\varphi''(X_x)]\, (X_x^2 - |x|); \tag{41}$$

the singleton $\{x = 0\}$ has Lebesgue measure zero and may be ignored. Substituting (41) and the formula for $\mathrm{Proj}_1^Y L_T^x(Y)$ into theorem 7.6 yields (40).

Strong measurability and Bochner integrability of the projected integrand follow from the operator statement of theorem 5.2 applied with $S = \mathrm{Proj}_{2,1}$, exactly as in the proof of theorem 7.6. □

8. A Stein bound at bidegree $(1,1)$ and a large-T obstruction

The kernel $K_T \in L^2(\mathbb{R}\times[0,T])$ of theorem 7.3 carries enough information to apply the Nourdin–Peccati fourth-moment machinery [10, 9] to the normalised projection $G_T \coloneqq \mathrm{Proj}_{1,1} F_T(\varphi)/\sigma_T$, where $\sigma_T^2 = \mathrm{Var}(\mathrm{Proj}_{1,1} F_T(\varphi)) = \|K_T\|^2_{L^2(\mathbb{R}\times[0,T])}$ by theorem 7.7 and theorem 7.4. The Stein bound obtained this way is non-asymptotic and explicit at fixed T (theorem 8.3). As $T \to \infty$ with $\varphi \in C_c^\infty(\mathbb{R})$ fixed, however, σ_T^2 and the relevant contraction norm $\|\tilde f_T \otimes_1 \tilde f_T\|_{\mathcal{H}\otimes\mathcal{H}}$ scale at the same rate in T, and the resulting bound does not vanish. A $T \to \infty$ CLT for G_T is therefore stated as a conjecture, with the estimate needed to close it (theorem 8.7).

Let $\mathcal{H} \coloneqq \mathcal{H}_X \oplus \mathcal{H}_Y$ be the orthogonal Hilbert sum, with $\mathcal{H}_X = L^2(\mathbb{R}, dx)$ and $\mathcal{H}_Y = L^2(\mathbb{R}_+, ds)$. The joint isonormal Gaussian process $W : \mathcal{H} \to L^2(\Omega)$ is defined on elementary $h_X \oplus h_Y$ by $W(h_X \oplus h_Y) \coloneqq W^X(h_X) + W^Y(h_Y)$; the chaoses $\mathcal{H}_m^X \otimes \mathcal{H}_n^Y \subseteq L^2(\Omega)$ are contained in the $(m+n)$-th joint chaos, and by independence the joint q-th multiple integral I_q restricted to $\mathcal{H}_X^{\otimes m} \otimes \mathcal{H}_Y^{\otimes n} \subseteq \mathcal{H}^{\otimes(m+n)}$ agrees with the product $I_m^X \otimes I_n^Y$ of the single-source multiple integrals.

We embed the bilinear kernel $K_T\colon \mathcal{H}_X \times \mathcal{H}_Y \to \mathbb{R}$ of (26) into $\mathcal{H}\otimes\mathcal{H}$ as the (non-symmetric) element f_T supported on the off-diagonal block $\mathcal{H}_X \otimes \mathcal{H}_Y$ with value K_T and zero on the other three blocks. Its symmetrisation

$$\tilde f_T \;\coloneqq\; \tfrac{1}{2}(f_T + f_T^\tau), \qquad f_T^\tau(\alpha,\beta) \coloneqq f_T(\beta,\alpha), \tag{42}$$

is supported on the two off-diagonal blocks $\mathcal{H}_X \otimes \mathcal{H}_Y$ and $\mathcal{H}_Y \otimes \mathcal{H}_X$, each carrying value $K_T/2$.

Lemma 8.1 (Joint chaos representation and isometry)**.** *With $f_T, \tilde f_T$ as in* (42)*,*

$$\|\tilde f_T\|^2_{\mathcal{H}\otimes\mathcal{H}} \;=\; \tfrac{1}{2}\, \|K_T\|^2_{L^2(\mathbb{R}\times[0,T])}, \qquad \mathrm{Proj}_{1,1} F_T(\varphi) \;=\; I_2(\tilde f_T), \tag{43}$$

and consequently $\mathbb{E}[I_2(\tilde f_T)^2] = 2\,\|\tilde f_T\|^2_{\mathcal{H}\otimes\mathcal{H}} = \|K_T\|^2_{L^2(\mathbb{R}\times[0,T])} = \sigma_T^2$, in agreement with theorem 7.7.

Proof. The norm identity is direct: the two off-diagonal blocks $\mathcal{H}_X \otimes \mathcal{H}_Y$ and $\mathcal{H}_Y \otimes \mathcal{H}_X$ each carry $K_T/2$ and the other two blocks are zero, so $\|\tilde f_T\|^2_{\mathcal{H}\otimes\mathcal{H}} = 2 \cdot (1/4)\, \|K_T\|^2_{L^2(\mathbb{R}\times[0,T])} = \frac{1}{2}\|K_T\|^2_{L^2(\mathbb{R}\times[0,T])}$.

For the identification $\mathrm{Proj}_{1,1} F_T(\varphi) = I_2(\tilde f_T)$, we first treat the case of elementary tensors. Fix $h \in \mathcal{H}_X$ and $g \in \mathcal{H}_Y$, and write $h \oplus 0$, $0 \oplus g \in \mathcal{H}$ for their embeddings as vectors in the orthogonal sum. The Wiener product formula on the joint isonormal $W : \mathcal{H} \to L^2(\Omega)$ [7, Prop. 1.1.2] gives, for $u, v \in \mathcal{H}$,

$$I_1(u)\, I_1(v) \;=\; I_2(u \otimes_s v) \;+\; \langle u, v\rangle_{\mathcal{H}}, \qquad u \otimes_s v \coloneqq \tfrac{1}{2}(u\otimes v + v \otimes u).$$

Applying this with $u = h \oplus 0$ and $v = 0 \oplus g$, the cross inner product vanishes, $\langle h \oplus 0,\, 0 \oplus g\rangle_{\mathcal{H}} = \langle h, 0\rangle_{\mathcal{H}_X} + \langle 0, g\rangle_{\mathcal{H}_Y} = 0$, and $I_1(h \oplus 0) = W^X(h)$, $I_1(0 \oplus g) = W^Y(g)$. Hence

$$W^X(h)\, W^Y(g) \;=\; I_2\big((h \oplus 0) \otimes_s (0 \oplus g)\big). \tag{44}$$

Now, by theorem 7.3, $\mathrm{Proj}_{1,1} F_T(\varphi) = \delta_X \delta_Y K_T$ is the iterated Skorokhod (double Wiener–Itô) integral of K_T. For an elementary kernel $K_T = h \otimes g$ with $h \in \mathcal{H}_X$, $g \in \mathcal{H}_Y$, this iterated integral is by definition $\delta_X \delta_Y (h \otimes g) = W^X(h)\, W^Y(g)$, which by (44) equals $I_2((h \oplus 0) \otimes_s (0 \oplus g))$. The symmetric tensor on the right is exactly the symmetrisation of the off-diagonal embedding of $h \otimes g$, i.e. $\tilde{f}_T$ in the case $K_T = h \otimes g$.

For general $K_T \in L^2(\mathbb{R} \times [0,T])$ both sides are bounded linear operators in K_T (the right-hand side via the L^2 continuity of I_2, the left-hand side via the L^2 continuity of $\delta_X \delta_Y$ on product step functions), and they agree on the dense linear span of elementary tensors $h \otimes g$, so they agree on all of $L^2(\mathbb{R} \times [0,T])$. This gives $\mathrm{Proj}_{1,1} F_T(\varphi) = I_2(\tilde{f}_T)$.

The final identity $\mathbb{E}[I_2(\tilde{f}_T)^2] = 2\, \|\tilde{f}_T\|^2_{\mathcal{H} \otimes \mathcal{H}}$ is the standard $q!\, \|\cdot\|^2$ isometry for the q-th multiple integral [7, Prop. 1.1.4], here at $q = 2$. □

The 1-contraction $\tilde{f}_T \otimes_1 \tilde{f}_T \in \mathcal{H} \otimes \mathcal{H}$ is defined by

$$(\tilde{f}_T \otimes_1 \tilde{f}_T)(\alpha, \beta) \;\coloneqq\; \int_{\mathcal{H}} \tilde{f}_T(\alpha, \gamma)\, \tilde{f}_T(\gamma, \beta)\, d\gamma, \qquad \alpha, \beta \in \mathcal{H}.$$

Proposition 8.2 (Contraction kernels)**.** *Define the spatial and temporal kernels*

$$G^X_T(s, s') \;\coloneqq\; \int_{\mathbb{R}} K_T(\xi, s)\, K_T(\xi, s')\, d\xi, \qquad s, s' \in [0,T], \tag{45}$$

$$G^Y_T(\xi, \xi') \;\coloneqq\; \int_0^T K_T(\xi, s)\, K_T(\xi', s)\, ds, \qquad \xi, \xi' \in \mathbb{R}. \tag{46}$$

Then $G^X_T \in L^2([0,T]^2)$ and $G^Y_T \in L^2(\mathbb{R}^2)$. On $\mathcal{H} \otimes \mathcal{H}$,

$$\tilde{f}_T \otimes_1 \tilde{f}_T\big|_{\mathcal{H}_X \otimes \mathcal{H}_X} \;=\; \tfrac{1}{4}\, G^Y_T, \qquad \tilde{f}_T \otimes_1 \tilde{f}_T\big|_{\mathcal{H}_Y \otimes \mathcal{H}_Y} \;=\; \tfrac{1}{4}\, G^X_T, \tag{47}$$

and the two off-diagonal blocks of $\tilde{f}_T \otimes_1 \tilde{f}_T$ vanish. Consequently

$$\|\tilde{f}_T \otimes_1 \tilde{f}_T\|^2_{\mathcal{H} \otimes \mathcal{H}} \;=\; \tfrac{1}{16}\, \big(\|G^X_T\|^2_{L^2([0,T]^2)} \,+\, \|G^Y_T\|^2_{L^2(\mathbb{R}^2)}\big). \tag{48}$$

Proof. Cauchy–Schwarz on (45) and (46) together with $K_T \in L^2(\mathbb{R} \times [0,T])$ give $G^X_T \in L^2([0,T]^2)$ and $G^Y_T \in L^2(\mathbb{R}^2)$. For (47), decompose the contraction integral $\int_{\mathcal{H}} \tilde{f}_T(\alpha, \gamma) \tilde{f}_T(\gamma, \beta)\, d\gamma$ as $\int_{\mathcal{H}_X} + \int_{\mathcal{H}_Y}$. For $\alpha, \beta \in \mathcal{H}_X$ the $\mathcal{H}_X$-integral vanishes because $\tilde{f}_T$ has no $\mathcal{H}_X \otimes \mathcal{H}_X$ block; the $\mathcal{H}_Y$-integral evaluates to

$$\int_0^T \tfrac{1}{2} K_T(\alpha, \eta)\, \tfrac{1}{2} K_T(\beta, \eta)\, d\eta \;=\; \tfrac{1}{4}\, G^Y_T(\alpha, \beta),$$

using that on $\mathcal{H}_X \otimes \mathcal{H}_Y$ and $\mathcal{H}_Y \otimes \mathcal{H}_X$ the symmetrised $\tilde{f}_T$ equals $K_T/2$. The case $\alpha, \beta \in \mathcal{H}_Y$ is symmetric. For $\alpha \in \mathcal{H}_X$, $\beta \in \mathcal{H}_Y$ both summands vanish (the $\mathcal{H}_X$-integral needs $\tilde{f}_T$ on $\mathcal{H}_Y \otimes \mathcal{H}_Y$ and the $\mathcal{H}_Y$-integral needs $\tilde{f}_T$ on $\mathcal{H}_X \otimes \mathcal{H}_X$), hence the off-diagonal blocks of $\tilde{f}_T \otimes_1 \tilde{f}_T$ are zero. Equation (48) follows by summing the squared L^2 norms over the two surviving blocks: each contributes $(1/4)^2 \|G^\bullet_T\|^2 = \frac{1}{16} \|G^\bullet_T\|^2$. □

Theorem 8.3 (Non-asymptotic Stein bound at bidegree $(1,1)$)**.** *Let $\varphi \in C^\infty_c(\mathbb{R})$ and $T > 0$ be such that $\sigma^2_T = \mathrm{Var}(\mathrm{Proj}_{1,1} F_T(\varphi)) > 0$. Set $G_T \coloneqq \mathrm{Proj}_{1,1} F_T(\varphi)/\sigma_T$. Then*

$$d_{\mathrm{TV}}(G_T,\, N(0,1)) \;\le\; \frac{4\sqrt{2}}{\sigma^2_T}\, \|\tilde{f}_T \otimes_1 \tilde{f}_T\|_{\mathcal{H} \otimes \mathcal{H}} \;=\; \frac{\sqrt{2}}{\sigma^2_T} \sqrt{\|G^X_T\|^2_{L^2([0,T]^2)} \,+\, \|G^Y_T\|^2_{L^2(\mathbb{R}^2)}}. \tag{49}$$

Both sides depend on φ and T only through K_T.

Proof. We use the fundamental Stein inequality from Malliavin calculus [9, Thm. 5.1.3]: for any centred $F \in \mathbb{D}^{1,2}$ with $\mathrm{Var}(F) = \sigma^2$,

$$d_{\mathrm{TV}}(F/\sigma,\, N(0,1)) \;\le\; \frac{2}{\sigma^2}\,\sqrt{\mathrm{Var}(\langle DF, -DL^{-1}F\rangle_{\mathcal{H}})}.$$

We compute the right-hand side explicitly for $F = I_2(\tilde f_T)$. The Ornstein–Uhlenbeck generator on the joint isonormal acts on the second Wiener chaos by $LF = -2F$, so $L^{-1}F = -F/2$ and $-DL^{-1}F = \frac{1}{2}DF$. Hence

$$\langle DF, -DL^{-1}F\rangle_{\mathcal{H}} \;=\; \tfrac{1}{2}\,\|DF\|^2_{\mathcal{H}}. \tag{50}$$

Next we expand $\|DF\|^2_{\mathcal{H}}$ for $F = I_2(\tilde f_T)$. The Malliavin derivative is $DF(\zeta) = 2\,I_1(\tilde f_T(\zeta,\cdot))$, $\zeta \in \mathcal{H}$, and the single-integral product formula $I_1(g)^2 = \|g\|^2 + I_2(g\otimes g)$ gives, for each ζ,

$$DF(\zeta)^2 \;=\; 4\,I_1(\tilde f_T(\zeta,\cdot))^2 \;=\; 4\,\|\tilde f_T(\zeta,\cdot)\|^2 \;+\; 4\,I_2\big(\tilde f_T(\zeta,\cdot)\otimes \tilde f_T(\zeta,\cdot)\big).$$

Integrating ζ over $\mathcal{H}$ and using $\int_{\mathcal{H}} \tilde f_T(\zeta,\cdot)\otimes\tilde f_T(\zeta,\cdot)\,d\zeta = \tilde f_T\otimes_1\tilde f_T$ together with $\int_{\mathcal{H}}\|\tilde f_T(\zeta,\cdot)\|^2\,d\zeta = \|\tilde f_T\|^2_{\mathcal{H}\otimes\mathcal{H}}$,

$$\|DF\|^2_{\mathcal{H}} \;=\; 4\,\|\tilde f_T\|^2_{\mathcal{H}\otimes\mathcal{H}} \;+\; 4\,I_2(\tilde f_T\otimes_1\tilde f_T) \;=\; 2\sigma_T^2 \;+\; 4\,I_2(\tilde f_T\otimes_1\tilde f_T),$$

the last equality using $\sigma_T^2 = 2\,\|\tilde f_T\|^2_{\mathcal{H}\otimes\mathcal{H}}$ from theorem 8.1. Combining with (50),

$$\langle DF, -DL^{-1}F\rangle_{\mathcal{H}} \;=\; \sigma_T^2 \;+\; 2\,I_2(\tilde f_T\otimes_1\tilde f_T), \qquad \mathbb{E}[\langle DF, -DL^{-1}F\rangle_{\mathcal{H}}] = \sigma_T^2. \tag{51}$$

The variance is therefore

$$\mathrm{Var}(\langle DF, -DL^{-1}F\rangle_{\mathcal{H}}) \;=\; 4\,\mathbb{E}[I_2(\tilde f_T\otimes_1\tilde f_T)^2] \;=\; 4\cdot 2\,\|\tilde f_T\otimes_1\tilde f_T\|^2_{\mathcal{H}\otimes\mathcal{H}} \;=\; 8\,\|\tilde f_T\otimes_1\tilde f_T\|^2_{\mathcal{H}\otimes\mathcal{H}}$$

(using that $\tilde f_T\otimes_1\tilde f_T$ is symmetric, by symmetry of $\tilde f_T$, so the isometry $\mathbb{E}[I_2(g)^2] = 2\|g\|^2$ applies). Substituting back yields

$$d_{\mathrm{TV}}(G_T,\, N(0,1)) \;\le\; \frac{2}{\sigma_T^2}\,\sqrt{8}\,\|\tilde f_T\otimes_1\tilde f_T\|_{\mathcal{H}\otimes\mathcal{H}} \;=\; \frac{4\sqrt{2}}{\sigma_T^2}\,\|\tilde f_T\otimes_1\tilde f_T\|_{\mathcal{H}\otimes\mathcal{H}},$$

which is the first inequality in (49). Substituting $\|\tilde f_T\otimes_1\tilde f_T\| = \frac{1}{4}\sqrt{\|G_T^X\|^2 + \|G_T^Y\|^2}$ from theorem 8.2 gives the equality. □

8.1. **Behaviour as $T\to\infty$.** The right-hand side of (49) depends on T only through $\|K_T\|^2$ and the L^2 norms of G_T^X, G_T^Y. Each of these admits a Brownian-time rescaling: with $x = u\sqrt{T}$ and $s = T\tau$, $\tau\in[0,1]$,

$$k_T(u\sqrt{T}, T\tau) \;=\; \mathrm{erf}\Big(\tfrac{u}{\sqrt{2\tau}}\Big) - \mathrm{erf}\Big(\tfrac{u}{\sqrt{2}}\Big) \;=:\; k(u,\tau), \tag{52}$$

which is T-independent. For $\varphi\in C_c^\infty(\mathbb{R})$ a Taylor expansion of $a(x) \coloneqq \mathbb{E}[\varphi'(X_x)]$ in inverse powers of $|x|$ gives, when the leading moment $\int y\,\varphi(y)\,dy \neq 0$,

$$a(x) \;=\; c_\varphi\,|x|^{-3/2} \;+\; O(|x|^{-5/2}), \qquad c_\varphi \;\coloneqq\; \frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}} y\,\varphi(y)\,dy, \qquad |x|\to\infty. \tag{53}$$

The expansion follows from Taylor-expanding the Gaussian density $(2\pi|x|)^{-1/2}\,e^{-y^2/(2|x|)}$ in inverse powers of $|x|$ and using compact support of φ: the leading term in the expansion vanishes because $\int_{\mathbb{R}}\varphi'(y)\,dy = 0$, and the $|x|^{-3/2}$ coefficient comes from $\int_{\mathbb{R}} y^2\,\varphi'(y)\,dy = -2\int_{\mathbb{R}} y\,\varphi(y)\,dy$ by integration by parts. The next term in the density expansion contributes $|x|^{-5/2}$ with coefficient $d_\varphi = -(2\sqrt{2\pi})^{-1}\int_{\mathbb{R}} y^3\,\varphi(y)\,dy$, which is generically nonzero when c_φ is.

Proposition 8.4 (Same-rate growth)**.** *Assume $\varphi \in C_c^\infty(\mathbb{R})$ satisfies* (53) *with $c_\varphi \neq 0$. There exist constants $I_J, I_X, I_Y \in (0,\infty)$, depending only on φ, such that as $T \to \infty$,*

$$\sigma_T^2 \;\sim\; c_\varphi^2\, I_J\, T, \qquad \|G_T^X\|^2_{L^2([0,T]^2)} \;\sim\; c_\varphi^4\, I_X\, T^2, \qquad \|G_T^Y\|^2_{L^2(\mathbb{R}^2)} \;\sim\; c_\varphi^4\, I_Y\, T^2. \tag{54}$$

Hence

$$\frac{\|\tilde f_T \otimes_1 \tilde f_T\|_{\mathcal{H}\otimes\mathcal{H}}}{\sigma_T^2} \;\xrightarrow[T\to\infty]{}\; \frac{1}{4}\,\frac{\sqrt{I_X + I_Y}}{I_J} \;\in\; (0,\infty). \tag{55}$$

Proof. We start with a boundary-layer estimate for the diagonal slice $\int_0^1 k(u,\tau)^2\,d\tau$ at $u \to 0$. The function $k(u,\tau) = \operatorname{erf}(u/\sqrt{2\tau}) - \operatorname{erf}(u/\sqrt{2})$ satisfies $k(0,\tau) = 0$ and $|k(u,\tau)| \le 2$ uniformly in u, τ. For $u \to 0^+$ we split the τ-integral at $\tau = u^2$. On $\tau \in (0, u^2]$, both $u/\sqrt{2\tau} \ge 1/\sqrt{2}$ and we use the crude bound $|k| \le 2$, giving a contribution $\le 4u^2$ to $\int_0^1 k(u,\tau)^2\,d\tau$. On $\tau \in (u^2, 1]$, the Taylor expansion $\operatorname{erf}(z) = (2z/\sqrt{\pi}) - (2z^3)/(3\sqrt{\pi}) + O(z^5)$ as $z \to 0$ gives

$$k(u,\tau) \;=\; u\sqrt{2/\pi}\,\Big(\frac{1}{\sqrt{\tau}} - 1\Big) \;+\; O\!\big(u^3\,\tau^{-3/2}\big),$$

uniformly for $\tau \ge u^2$, and hence

$$\int_{u^2}^1 k(u,\tau)^2\,d\tau \;=\; \tfrac{2u^2}{\pi}\int_{u^2}^1 \Big(\tfrac{1}{\sqrt{\tau}} - 1\Big)^2\,d\tau \;+\; O(u^2).$$

The leading integral is $\int_{u^2}^1 (\tau^{-1/2} - 1)^2\,d\tau = \int_{u^2}^1 (\tau^{-1} - 2\tau^{-1/2} + 1)\,d\tau = [-\log\tau\,]_{u^2}^1 - 4[\sqrt{\tau}\,]_{u^2}^1 + (1 - u^2) = -2\log u - 4 + 4u + 1 - u^2$, so

$$\int_0^1 k(u,\tau)^2\,d\tau \;=\; -\tfrac{4}{\pi}\,u^2\log u \;+\; O(u^2) \qquad (u \to 0^+), \tag{56}$$

and the symmetric estimate holds for $u \to 0^-$. In particular $\int_0^1 k(u,\tau)^2\,d\tau \to 0$ as $u \to 0$.

We next derive the scaling of σ_T^2 and the finiteness of I_J. Write

$$\sigma_T^2 \;=\; \int_{\mathbb{R}^2} a(x)a(x')\,\langle \sigma_x, \sigma_{x'}\rangle\, J(x,x',T)\,dx\,dx', \qquad J(x,x',T) \coloneqq \int_0^T k_T(x,s)k_T(x',s)\,ds.$$

Substitute $x = u\sqrt{T}$, $x' = u'\sqrt{T}$, $s = T\tau$. By (52), $J(u\sqrt{T}, u'\sqrt{T}, T) = T\,\tilde J(u,u')$, where

$$\tilde J(u,u') \;\coloneqq\; \int_0^1 k(u,\tau)\,k(u',\tau)\,d\tau.$$

The inner product satisfies $\langle \sigma_{u\sqrt{T}}, \sigma_{u'\sqrt{T}}\rangle_{\mathcal{H}_X} = \sqrt{T}\,\min(|u|,|u'|)\,\mathbf{1}_{uu'>0}$, and $dx\,dx' = T\,du\,du'$. By (53), $a(u\sqrt{T}) = c_\varphi\,(|u|\sqrt{T})^{-3/2} + r(u\sqrt{T})$ with $|r(x)| \le C|x|^{-5/2}$ for $|x| \ge 1$ and $|a(x)|$ bounded globally. Substituting the leading term and collecting powers of T gives

$$\sigma_T^2 \;=\; c_\varphi^2\,T\,I_J \;+\; R_\sigma(T), \quad I_J \;\coloneqq\; \int_{\mathbb{R}^2} |uu'|^{-3/2}\,\min(|u|,|u'|)\,\mathbf{1}_{uu'>0}\,\tilde J(u,u')\,du\,du', \tag{57}$$

with $R_\sigma(T)$ the contribution of the $r(x)\,r(x')$ and cross terms (estimated in the correction analysis below). We show $I_J \in (0,\infty)$ by separate estimates at the origin and at infinity.

Near the origin, by (56) and Cauchy–Schwarz,

$$|\tilde J(u,u')| \;\le\; \sqrt{\tilde J(u,u)\,\tilde J(u',u')} \;\le\; C\,|u|\,|u'|\,\sqrt{|\log u|\,|\log u'|}$$

for $|u|, |u'| \le 1/2$ with $uu' > 0$. The integrand of I_J is then bounded above by

$$\begin{aligned} &|uu'|^{-3/2}\,\min(|u|,|u'|)\,C\,|uu'|\,\sqrt{|\log u|\,|\log u'|} \\ &\quad=\; C\,\frac{\min(|u|,|u'|)}{|uu'|^{1/2}}\,\sqrt{|\log u|\,|\log u'|} \;\le\; C\,\sqrt{|\log u|\,|\log u'|}, \end{aligned}$$

using $\min(|u|,|u'|) \le \sqrt{|u||u'|}$. Integrating $|\log u|\,|\log u'|$ over $[0,1/2]^2$ is finite, so the integrand is integrable near the origin (the opposite-sign quadrant contributes 0 by the indicator $\mathbf{1}_{uu'>0}$).

At infinity, for $z \to \infty$ the standard tail estimate $1-\operatorname{erf}(z) \sim e^{-z^2}/(z\sqrt{\pi})$ gives, for $u \to +\infty$ and $\tau \in [0,1)$,

$$k(u,\tau) \;=\; \frac{1}{u\sqrt{\pi}}\big(\sqrt{2}\,e^{-u^2/2} - \sqrt{2\tau}\,e^{-u^2/(2\tau)}\big)\,(1+o(1)) \;=\; O\big(u^{-1}\,e^{-u^2/2}\big),$$

uniformly for $\tau \le 1-\delta$. On the strip $\tau \in [1-\delta,1]$, $k(u,\tau)$ obeys the same bound by Taylor expansion in τ around $\tau = 1$. Hence

$$\tilde{J}(u,u') \;\le\; \sqrt{\tilde{J}(u,u)\,\tilde{J}(u',u')} \;\le\; C\,(|u||u'|)^{-1}\,e^{-(u^2+u'^2)/2}$$

for $|u|,|u'| \ge 1$ with $uu' > 0$. The integrand of I_J is then bounded by

$$|uu'|^{-3/2}\,\min(|u|,|u'|)\,C\,(|u||u'|)^{-1}\,e^{-(u^2+u'^2)/2} \;\le\; C\,\min(|u|,|u'|)\,(|u||u'|)^{-5/2}\,e^{-(u^2+u'^2)/2},$$

integrable on $\mathbb{R}^2$. Hence $I_J \in (0,\infty)$, and the same-sign quadrant gives a strictly positive contribution provided $c_\varphi \neq 0$, so $I_J > 0$.

By Cauchy–Schwarz on (45)–(46),

$$\|G_T^Y\|^2_{L^2(\mathbb{R}^2)} \;=\; \int_{\mathbb{R}^4} a(x_1)a(x_2)a(x_3)a(x_4)\,\langle\sigma_{x_1},\sigma_{x_3}\rangle\,\langle\sigma_{x_2},\sigma_{x_4}\rangle \times\; J(x_1,x_2,T)\,J(x_3,x_4,T)\,dx_1\,dx_2\,dx_3\,dx_4.$$

Under $x_i = u_i\sqrt{T}$ for $i = 1,2,3,4$, the substitution contributes:

- four factors $a(x_i) = c_\varphi(|u_i|\sqrt{T})^{-3/2} + \tilde{r}(u_i\sqrt{T})$, each giving $T^{-3/4}$ from the leading term, for a total of T^{-3};
- two inner products $\langle\sigma_{x_i},\sigma_{x_j}\rangle = \sqrt{T}\,\min(|u_i|,|u_j|)\,\mathbf{1}_{u_iu_j>0}$, contributing T;
- two factors $J(x_i,x_j,T) = T\,\tilde{J}(u_i,u_j)$, contributing T^2;
- four Jacobians $dx_i = \sqrt{T}\,du_i$, contributing T^2.

The leading power is $T^{-3}\cdot T\cdot T^2\cdot T^2 = T^2$, giving

$$\|G_T^Y\|^2_{L^2(\mathbb{R}^2)} \;=\; c_\varphi^4\,T^2\,I_Y \;+\; R_Y(T), \tag{58}$$

where

$$I_Y \;:=\; \int_{\mathbb{R}^4} |u_1u_2u_3u_4|^{-3/2}\,\min(|u_1|,|u_3|)\,\mathbf{1}_{u_1u_3>0}\,\min(|u_2|,|u_4|)\,\mathbf{1}_{u_2u_4>0} \times\; \tilde{J}(u_1,u_2)\,\tilde{J}(u_3,u_4)\,du_1\,du_2\,du_3\,du_4. \tag{59}$$

Finiteness of I_Y follows from $|\tilde{J}(u,u')| \le \tilde{J}(u,u)^{1/2}\tilde{J}(u',u')^{1/2}$ and the boundary-layer and exponential-decay estimates above. For the remainder $R_Y(T)$: each replacement of one of the four $c_\varphi(|u_i|\sqrt{T})^{-3/2}$ factors by $\tilde{r}(u_i\sqrt{T}) = O((|u_i|\sqrt{T})^{-5/2})$ on $\{|u_i|\sqrt{T}\ge 1\}$ loses an extra factor of $T^{-1/2}$, so a single replacement gives $T^{3/2} = o(T^2)$, and two or more replacements give T^1 or smaller. For the small-$|x_i|$ contributions, partition $\mathbb{R}^4$ by the four indicators $\mathbf{1}_{|x_i|\ge 1}$. The all-large region $\{|x_i|\ge 1 : i = 1,\dots,4\}$ gives the leading $c_\varphi^4\,T^2\,I_Y$; each of the remaining fifteen regions contains at least one $|x_j| < 1$. Consider the representative region $\{|x_1| < 1,\ |x_2|,|x_3|,|x_4| \ge 1\}$. The constraint restricts u_1 to a Lebesgue-measure-$O(T^{-1/2})$ set; $|a(x_1)| \le \|\varphi'\|_\infty$ replaces the leading $T^{-3/4}$ factor; the bound $|\langle\sigma_{x_1},\sigma_{x_3}\rangle| \le \sqrt{T}\,|u_1|$ holds when $|u_1| < 1/\sqrt{T} \le |u_3|$ (so the min equals $|u_1|$); and the boundary-layer estimate

$|k_T(x,s)| \le C|x|/\sqrt{s}$ on $\{|x| \le 1,\ s \ge x^2\}$ gives $|J(x_1,x_2,T)| \le C\,|u_1|\,T$ in place of the full size T. Collecting,

$$T^{-9/4}\cdot\sqrt{T}\cdot\sqrt{T}\cdot T\cdot T \;=\; T^{3/4}, \qquad \int_{|u_1|<1/\sqrt{T}} |u_1|^2\,du_1 \;=\; \tfrac{2}{3}\,T^{-3/2},$$

and the remaining (u_2,u_3,u_4) integral converges. With the Jacobian $dx_1\cdots dx_4 = T^2\,du_1\cdots du_4$, the region contributes $T^2\cdot T^{3/4}\cdot T^{-3/2} = T^{5/4} = o(T^2)$. The other fourteen small-$|x_i|$ regions are bounded analogously; each additional small variable yields a further factor of $|u_i|$ and a further measure-shrinking $O(T^{-1/2})$, so all of them remain $o(T^2)$. Hence $R_Y(T) = o(T^2)$.

The symmetric analysis for $\|G_T^X\|^2_{L^2([0,T]^2)}$ uses the substitution $s_i = T\tau_i$ in the four-fold integral defining $\|G_T^X\|^2$. The inner product $\langle \sigma_{x_1}, \sigma_{x_2}\rangle$ pairs with the four-fold integrand differently in this case: rewriting

$$G_T^X(s,s') \;=\; \int_{\mathbb{R}^2} a(x)a(x')\,\langle \sigma_x, \sigma_{x'}\rangle\, k_T(x,s)\,k_T(x',s')\,dx\,dx',$$

substituting $x = u\sqrt{T}$, $x' = u'\sqrt{T}$, $s_i = T\tau_i$ and collecting powers gives

$$\|G_T^X\|^2_{L^2([0,T]^2)} \;=\; c_\varphi^4\,T^2\,I_X \;+\; R_X(T), \tag{60}$$

where

$$I_X \;\coloneqq\; \int_{[0,1]^2}\int_{\mathbb{R}^4} |u_1u_2u_3u_4|^{-3/2}\,\min(|u_1|,|u_2|)\,\mathbf{1}_{u_1u_2>0}\,\min(|u_3|,|u_4|)\,\mathbf{1}_{u_3u_4>0} \\ \times\; k(u_1,\tau)\,k(u_2,\tau')\,k(u_3,\tau)\,k(u_4,\tau')\,du_1\,du_2\,du_3\,du_4\,d\tau\,d\tau'. \tag{61}$$

For finiteness of I_X, apply Cauchy–Schwarz in τ (pairing $k(u_1,\tau)$ with $k(u_3,\tau)$) and in τ' (pairing $k(u_2,\tau')$ with $k(u_4,\tau')$):

$$\int_0^1 k(u_1,\tau)\,k(u_3,\tau)\,d\tau \;\le\; \sqrt{\tilde J(u_1,u_1)\,\tilde J(u_3,u_3)},$$

and the analogous bound in τ'. The integrand of I_X is therefore bounded above by

$$|u_1u_2u_3u_4|^{-3/2}\,\min(|u_1|,|u_2|)\,\mathbf{1}_{u_1u_2>0}\,\min(|u_3|,|u_4|)\,\mathbf{1}_{u_3u_4>0} \\ \times\sqrt{\tilde J(u_1,u_1)\,\tilde J(u_3,u_3)\,\tilde J(u_2,u_2)\,\tilde J(u_4,u_4)}.$$

By (56), $\tilde J(u,u) = O(u^2|\log u|)$ as $u\to 0$, while the tail estimate of the previous paragraph gives $\tilde J(u,u) = O(u^{-2}\,e^{-u^2})$ as $|u|\to\infty$. Hence $\sqrt{\tilde J(u_i,u_i)} = O(|u_i|\sqrt{|\log u_i|})$ at $u_i\to 0$ and $O(|u_i|^{-1}\,e^{-u_i^2/2})$ at $|u_i|\to\infty$. The four-fold product is then integrable at the origin (each $|u_i|^{-3/2}\cdot|u_i|\sqrt{|\log u_i|}$ combined with the min indicators is locally integrable, as in the I_J argument) and at infinity (Gaussian decay in each u_i). Hence $I_X < \infty$.

The analogous statement for I_Y follows by exchanging the roles of the spatial and temporal variables: applying Cauchy–Schwarz in the two pairs (x_1,x_3) and (x_2,x_4) of (59) and using the same boundary-layer and exponential-decay estimates on the resulting $\tilde J(u_i,u_i)$ factors gives $I_Y < \infty$.

Finally, I_J, I_X, I_Y are strictly positive. For $u>0$ and $\tau\in(0,1)$, $u/\sqrt{2\tau} > u/\sqrt{2}$ and erf is strictly increasing, so $k(u,\tau) > 0$. Hence $\tilde J(u,u') = \int_0^1 k(u,\tau)\,k(u',\tau)\,d\tau > 0$ whenever $u,u' > 0$, and similarly for $u,u' < 0$ by oddness of k in u. The integrand of I_J is therefore non-negative on $\mathbb{R}^2$ and strictly positive on each same-sign quadrant $\{u,u'>0\}$, $\{u,u'<0\}$, each of positive Lebesgue measure on the set where the integrand is finite. So $I_J > 0$. For I_X and I_Y, the integrands defined in (61) and (59) contain only products of $k(u_i,\tau)$ or $k(u_i,\tau')$ with non-negative weights and indicators $\mathbf{1}_{u_iu_j>0}$. On the all-positive orthant $\{u_1,u_2,u_3,u_4>0\}$,

every indicator equals 1 and every factor $k(u_i,\tau)$, $k(u_i,\tau')$ is strictly positive for $(\tau,\tau')\in(0,1)^2$. The orthant has positive Lebesgue measure and the integrand is finite on a subset of positive measure, e.g. the bounded box $(0,M]^4\times(\delta,1)^2$ for any $M<\infty$ and $\delta>0$. Hence $I_X, I_Y>0$.

To control the remainder, observe that $a(x)=\mathbb{E}[\varphi'(X_x)]$ satisfies $|a(x)|\le\|\varphi'\|_\infty$ uniformly on $\mathbb{R}$, while the asymptotic expansion (53) provides the sharper estimate $|a(x)-c_\varphi|x|^{-3/2}|\le C\,|x|^{-5/2}$ for $|x|\ge 1$ only. The leading term in the substitution $x=u\sqrt{T}$ therefore applies on $\{|x|\ge 1\}$, i.e. on $\{|u|\ge 1/\sqrt{T}\}$. Define

$$\tilde r(x)\;:=\;a(x)-c_\varphi\,|x|^{-3/2}\,\mathbf{1}_{|x|\ge 1},$$

so that $|\tilde r(x)|\le\|\varphi'\|_\infty$ for $|x|<1$ and $|\tilde r(x)|\le C\,|x|^{-5/2}$ for $|x|\ge 1$. We split each spatial integral into the contributions from $\{|x|,|x'|\ge 1\}$, the mixed region $\{|x|\ge 1,\,|x'|<1\}$, and $\{|x|,|x'|<1\}$.

On $\{|x|,|x'|\ge 1\}$, substituting $a=c_\varphi|x|^{-3/2}+\tilde r$ produces three terms beyond the leading $c_\varphi^2 T\,I_J$ of (57). The cross term carries one factor of $\tilde r$ of size $|x|^{-5/2}$; under $x=u\sqrt{T}$, $x'=u'\sqrt{T}$ it collects powers $T^{-3/4}\cdot T^{-5/4}\cdot\sqrt{T}\cdot T\cdot T=T^{1/2}$ (one $T^{-3/4}$ from the leading factor, one $T^{-5/4}$ from $\tilde r$, $\sqrt{T}$ from $\langle\sigma\rangle$, T from J, T from $du\,du'$), so contributes $O(\sqrt{T})=o(T)$. The double-correction term carries two $\tilde r$ factors and contributes $T^{-5/4}\cdot T^{-5/4}\cdot\sqrt{T}\cdot T\cdot T=T^0=o(T)$.

On the mixed region $\{|x|\ge 1,\,|x'|<1\}$ (and symmetrically), the factor $a(x')$ is bounded by $\|\varphi'\|_\infty$ rather than by $c_\varphi|x'|^{-3/2}$, and $\langle\sigma_x,\sigma_{x'}\rangle=\min(|x|,|x'|)\,\mathbf{1}_{xx'>0}\le 1$. The boundary-layer estimate (56) extends to $|k_T(x',s)|\le C|x'|/\sqrt{s}$ for $|x'|\le 1$ and $s\ge|x'|^2$, hence $|J(x,x',T)|\le C\,|x'|\,\sqrt{T}$ on this region. The contribution is

$$\|\varphi'\|_\infty\Big(\int_{|x|\ge1}c_\varphi|x|^{-3/2}\,dx\Big)\Big(\int_{|x'|<1}|x'|\,dx'\Big)\cdot C\sqrt{T}\;=\;O(\sqrt{T})\;=\;o(T).$$

On $\{|x|,|x'|<1\}$, $|a(x)|,|a(x')|\le\|\varphi'\|_\infty$, the inner product is bounded by 1, and the boundary-layer estimate gives $|J(x,x',T)|\le C(1+|xx'|\log T)$ for $|x|,|x'|<1$; the contribution is $O(\log T)=o(T)$.

Collecting, $\sigma_T^2=c_\varphi^2\,T\,I_J+o(T)$. The analysis for $\|G_T^X\|^2$ proceeds the same way: each of the four $c_\varphi(|u_i|\sqrt{T})^{-3/2}$ factors splits as $c_\varphi(|u_i|\sqrt{T})^{-3/2}+\tilde r(u_i\sqrt{T})$; on $\{|u_i|\sqrt{T}\ge 1\}$ each replacement by $\tilde r$ loses $T^{-1/2}$ from the $|\tilde r|\le C|x|^{-5/2}$ bound, giving $T^2\cdot T^{-1/2}=T^{3/2}=o(T^2)$ after one replacement; two or more replacements give T^1 or smaller. The small-$|x_i|$ regions contribute $o(T^2)$ via the same boundary-layer bound $|k_T(x,s)|\le C|x|/\sqrt{s}$ on $\{|x|\le 1,\,s\ge x^2\}$ used in the $R_Y(T)$ analysis. Hence $R_X(T)=o(T^2)$.

Combining (57), (58), (60) with the $o(\cdot)$ remainders gives (54). The ratio (55) is then $\big(\frac{1}{16}\,c_\varphi^4(I_X+I_Y)\,T^2\big)^{1/2}/(c_\varphi^2\,I_J\,T)=\frac{1}{4}\,\sqrt{I_X+I_Y}/I_J\in(0,\infty)$, using (48). □

Corollary 8.5 (The bound does not vanish as $T\to\infty$)**.** *Under the hypotheses of theorem 8.4, the Stein bound* (49) *satisfies*

$$\liminf_{T\to\infty}\,d_{\mathrm{TV}}(G_T,\,N(0,1))_{\mathit{Stein}}\;>\;0,$$

where the subscript denotes the upper bound from theorem 8.3 rather than the actual d_{TV}. In particular, this bound alone does not yield a CLT for G_T as $T\to\infty$.

Remark 8.6. Theorem 8.5 is a statement about the upper bound in (49), not about G_T itself: the inequality is one-sided and could be loose. Whether G_T converges in distribution to $N(0,1)$ as $T\to\infty$ remains open, and numerical investigation of G_T is left for future work; the negative result of theorem 8.5 does not rule out the possibility that the bound is asymptotically loose and that G_T is nevertheless asymptotically Gaussian.

Conjecture 8.7 (Quantitative CLT for the bidegree-$(1,1)$ projection)**.** *For a suitable class of test functions φ (in particular requiring decay of $a(x)$ strictly faster than $|x|^{-3/2}$, or equivalently the vanishing of the leading moment $\int y\,\varphi(y)\,dy$), there exist a normalisation $\rho_T > 0$ and an exponent $\alpha > 0$ such that $\mathrm{Proj}_{1,1}F_T(\varphi)/\rho_T \to N(0,1)$ in total variation as $T \to \infty$, with rate $O(T^{-\alpha})$. The identification of the correct ρ_T and α in this limit, and a proof either under sharper contraction estimates than those of theorem 8.2 or under an alternative scaling regime (e.g. $\varphi_T(x) = \varphi(x/\sqrt{T})$), is left for future work.*

9. Conclusion

The product Wiener chaos decomposition $L^2(\Omega) = \bigoplus_{m,n\geq 0} \mathcal{H}_m^X \otimes \mathcal{H}_n^Y$ and the underlying Malliavin calculus on each Gaussian factor are standard. We use the IBM occupation-time factorisation $F_T(\varphi) = \int_{\mathbb{R}} \varphi(X_x)\, L_T^x(Y)\, dx$ to compute and interpret source-resolved bidegrees; the source projection formula (theorem 7.6) makes this possible. The centred functional splits orthogonally as

$$F - \mathbb{E}[F] \;=\; F_X^{\mathrm{pure}} \;+\; F_Y^{\mathrm{pure}} \;+\; F_{\mathrm{mix}},$$

with $F_X^{\mathrm{pure}} \in \bigoplus_{m\geq 1} \mathcal{H}_m^X \otimes \mathcal{H}_0^Y$, $F_Y^{\mathrm{pure}} \in \bigoplus_{n\geq 1} \mathcal{H}_0^X \otimes \mathcal{H}_n^Y$, and $F_{\mathrm{mix}} \in \bigoplus_{m,n\geq 1} \mathcal{H}_m^X \otimes \mathcal{H}_n^Y$. Each sector is detected by a partial Malliavin derivative: D_X together with the $1/m$-multiplier $\Pi_X = -L_X^{-1}$ recovers F_X^{pure} via the sourcewise Stein/Meyer identity $F - \mathbb{E}_X F = \delta_X(D_X\, \Pi_X F)$; symmetrically for F_Y^{pure}; and $D_X D_Y$ detects the mixed sector chaos-by-chaos, with $F_{1,1} = \delta_X\delta_Y\, \mathbb{E}[D_X D_Y F]$. We emphasised in theorem 4.8 that the naive sum $\delta_X(D_X \Pi_X F) + \delta_Y(D_Y \Pi_Y F)$ counts the mixed sector twice.

The occupation functional

$$F_T(\varphi) = \int_0^T \varphi(X_{Y_s})\, ds = \int_{\mathbb{R}} \varphi(X_x)\, L_T^x(Y)\, dx$$

is the canonical example: the right-hand integral makes the source factorization manifest. We computed the scenery derivative $D_X F_T$ directly; we treated the clock derivative through a smooth scenery X^ε, where $D_Y F_T^\varepsilon(s) = \int_s^T \varphi'(X^\varepsilon_{Y_u})\, \partial_x X^\varepsilon_{Y_u}\, du$ is standard, and we discussed the $\varepsilon \to 0$ limit as distributional / local-time-based; and we identified the bidegree-$(1,1)$ mixed component by direct product projection (theorem 7.3). The partial Malliavin operators D_X and D_Y lower the two source indices and thereby provide the bidegree language in which $(1,1)$ is identified as the first mixed scenery–clock component; the identification of the $(1,1)$-kernel itself comes from product projection, not from $D_X D_Y$. The first-order source split $\mathrm{Proj}_{1,0}F_T + \mathrm{Proj}_{0,1}F_T$ produces a sourcewise variance attribution that the total first-chaos projection does not.

The product chaos decomposition on $L^2(\Omega_X \times \Omega_Y)$ is the standard one, read here with respect to the finer source-resolved bidegree rather than introduced anew for IBM. The operators $D_X, D_Y, \delta_X, \delta_Y$ are the Malliavin derivative and Skorokhod integral applied to one Gaussian factor at a time; their role here is to organise the bidegree-(m,n) decomposition of the IBM occupation functional and to record the sectorwise Stein/Meyer bookkeeping with its double-counting correction. The source-resolved viewpoint suggests a probabilistic interpretation of the fourth-order/local-time traces of Allouba–Zheng [1] and related work, schematically with the pure-scenery sector $(m,0)$ contributing spatial second-derivative behaviour, the pure-clock sector $(0,n)$ contributing averaged clock moments, and the mixed sector (m,n), $m,n \geq 1$, carrying genuine scenery–clock coupling; we do not establish a PDE here, and that direction is left for future work.

The IBM occupation-time factorisation makes the product-chaos bidegree (m, n) computable; partial Malliavin derivatives on the two Gaussian factors record the resulting sector decomposition. The same source-by-source reading applies to other composite Gaussian functionals where the two sources play distinct roles.

The interaction-energy identity (theorem 7.7) underwrites a family of source-resolved control variates for Monte Carlo simulation of Brownian-time occupation functionals; an explicit numerical implementation and the associated variance budget are left to future work.

Acknowledgements. AI tools were used as interactive assistants for drafting, reorganizing, and generating mathematical arguments, under the author's direction. The author is responsible for all mathematical content and any errors.

References

[1] H. Allouba and W. Zheng, *Brownian-time processes: the PDE connection and the half-derivative generator*, Ann. Probab. **29** (2001), no. 4, 1780–1795.
[2] A. N. Borodin and P. Salminen, *Handbook of Brownian Motion — Facts and Formulae*, 2nd ed., Probability and Its Applications, Birkhäuser Verlag, Basel, 2002.
[3] Y. Hu and D. Nualart, *Renormalized self-intersection local time for fractional Brownian motion*, Ann. Probab. **33** (2005), no. 3, 948–983.
[4] Y. Hu and B. Øksendal, *Chaos expansion of local time of fractional Brownian motions*, Stochastic Anal. Appl. **20** (2002), no. 4, 815–837.
[5] S. Janson, *Gaussian Hilbert Spaces*, Cambridge Tracts in Mathematics, vol. 129, Cambridge University Press, Cambridge, 1997.
[6] D. Khoshnevisan and T. M. Lewis, *Stochastic calculus for Brownian motion on a Brownian fracture*, Ann. Appl. Probab. **9** (1999), no. 3, 629–667.
[7] D. Nualart, *The Malliavin Calculus and Related Topics*, 2nd ed., Probability and Its Applications, Springer-Verlag, Berlin, 2006.
[8] I. Nourdin and G. Peccati, *Weighted power variations of iterated Brownian motion*, Electron. J. Probab. **13** (2008), no. 43, 1229–1256.
[9] I. Nourdin and G. Peccati, *Normal Approximations with Malliavin Calculus: From Stein's Method to Universality*, Cambridge Tracts in Mathematics, vol. 192, Cambridge University Press, Cambridge, 2012.
[10] D. Nualart and G. Peccati, *Central limit theorems for sequences of multiple stochastic integrals*, Ann. Probab. **33** (2005), no. 1, 177–193.
[11] D. Nualart and J. Vives, *Chaos expansions and local times*, Publ. Mat. **36** (1992), no. 2B, 827–836.
[12] D. Revuz and M. Yor, *Continuous Martingales and Brownian Motion*, 3rd ed., Grundlehren der Mathematischen Wissenschaften, vol. 293, Springer-Verlag, Berlin, 1999.

Quijotic Research
Email address: ramirofontes@gmail.com
URL: www.quijotic.com